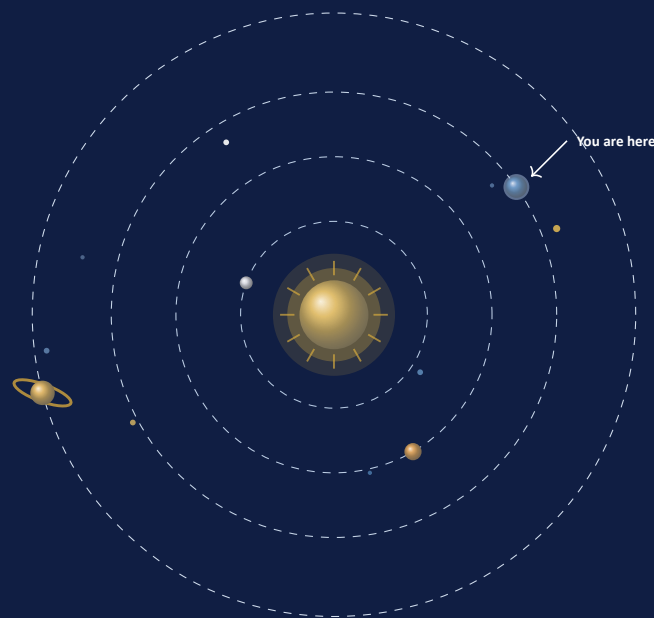


A JOURNEY THROUGH THE UNIVERSE

Astronomy

For Young Explorers

Ages 8–14



"Somewhere, something incredible is waiting to be known."

— Carl Sagan

Discover planets and stars, rockets and galaxies,
the Moon and the Sun, and your place in the universe.

Tawhid Bin Omar

Josephite Math Club • Version 1.0 • 25th July 2026

A Journey Through the Universe

Astronomy for Young Explorers

★ EX LIBRIS ★

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To Whom It May Concern

This is a humble compilation I pieced together for my enthusiastic and ambitious juniors — a small bridge between curiosity and the cosmos. May it spark in you the same wonder I feel every time I look up at the night sky. I hope it inspires you to love space, the universe, and the beautiful physics that governs it all.

Special thanks to
**Debbrota Kumar Biswas, Anthony Prince Costa, Animesh Saha, and
Tasmin Jahan**
for believing in my dreams and supporting my journey through
physics and problem solving.

— *Tawhid Bin Omar*



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PART I

Great discoveries begin with a single question: What is out there?

HAVE you ever looked up at the night sky and wondered what is out there? Those tiny twinkling lights. That big bright Moon. The dark space between them. Where does space begin? What lives there? And where do we fit in?

What is space, and where is Earth in the universe?

1.1

What Is Space?

Space begins about 100 kilometres above Earth. That is higher than any airplane flies! It is dark, cold, silent, and mostly empty. But space is also full of amazing things: stars, planets, moons, asteroids, comets, and galaxies.

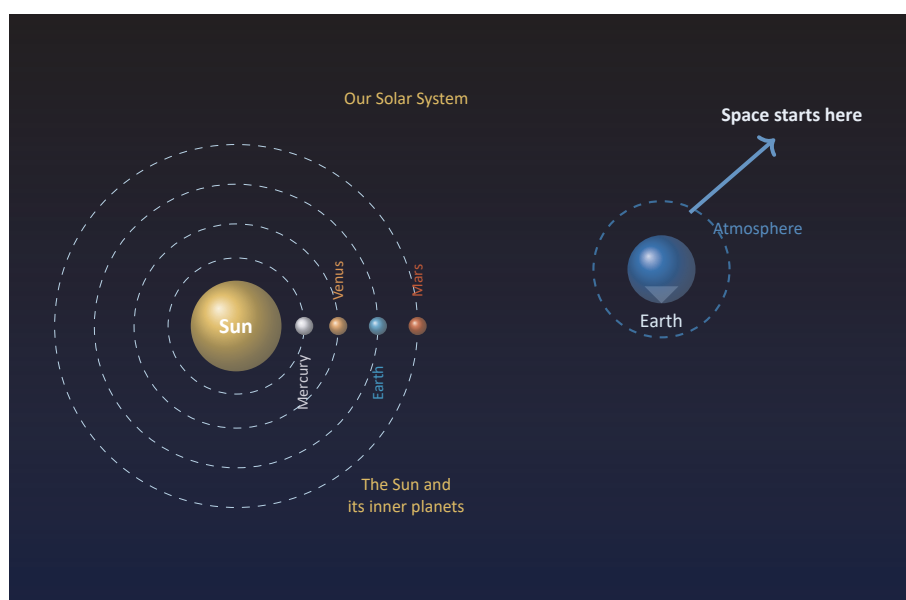


Figure 1.1: Earth is part of a solar system orbiting the Sun. Space begins just above our atmosphere.

Space

Space is the huge, mostly empty area beyond Earth's atmosphere. It contains everything: planets, stars, galaxies, and all the matter in the universe.

One Hour to Space

If you could drive a car straight up at 100 kilometres per hour, it would take only one hour to reach the edge of space!

1.2**We Live on a Planet**

Earth is our home. It is a planet — a large round world that orbits the Sun. Earth is just right for life. It has liquid water, breathable air, and a temperature that is not too hot and not too cold.



Figure 1.2: Earth from space — the Blue Marble.

- Earth is the **third planet** from the Sun.
- It takes **365 days** (one year) to travel entirely around the Sun.
- About **71%** of Earth's surface is covered with water.
- Earth is the only planet we know of that supports life.

Invisible Shield

Earth has a powerful magnetic field that protects us from the Sun's harmful particles. It acts just like an invisible force field!

1.3**The Sun — Our Nearest Star**

Every morning, the Sun rises and lights up the world. But what exactly is the Sun? It is a star — a huge ball of hot, glowing gas. The Sun is the closest star to Earth, and it gives us the light and warmth we need to survive.

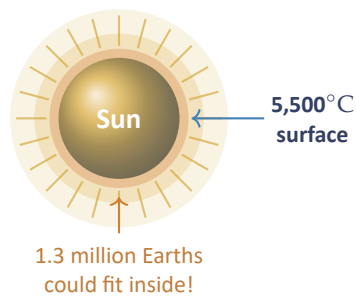


Figure 1.3: The Sun is enormous — more than a million Earths could fit inside it.

8-Minute Travel

Light from the Sun takes about 8 minutes to reach Earth. When you look at the Sun, you are actually seeing it as it was 8 minutes ago!

Make a Shadow Clock (Sundial)

You can track the Sun's motion using shadows! Take a stick (about 30 cm long) and push it into the ground outdoors in a sunny spot. Every hour, mark where the stick's shadow falls and write the time next to it. By the end of the day, you will have built a working sundial.

Observe: Why does the shadow move? Why is it shortest at noon? What happens to the shadow on a cloudy day?

The Sun Is a Star

All stars are giant balls of hot gas. Our Sun is a medium-sized yellow star. There are stars much bigger, much smaller, much hotter, and much cooler in the universe. At night, the stars you see are just other suns — they are just very, very far away.

Never Look at the Sun

The Sun is so incredibly bright that looking at it can hurt your eyes forever. Never look directly at the Sun, even with sunglasses. Always use special solar filters or eclipse glasses.

1.4

Earth's Neighbour: The Moon

The Moon is Earth's closest neighbour in space. It orbits our planet once every 27.3 days. The Moon does not make its own light — it shines because sunlight bounces off its surface, just like a mirror.

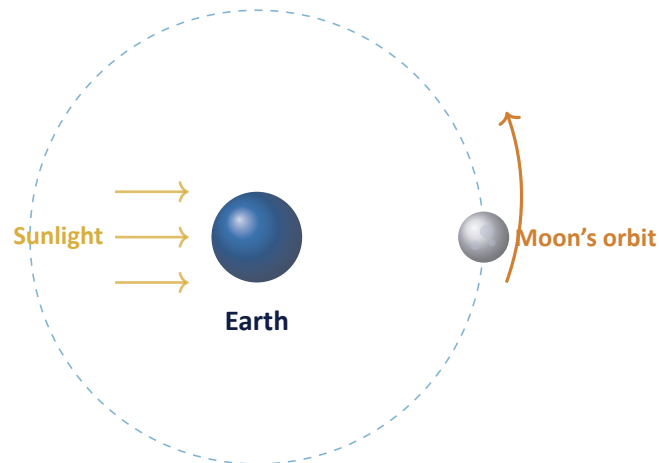


Figure 1.4: The Moon orbits Earth and shines by reflecting sunlight.

Always the Same Face

The Moon spins at exactly the same rate that it orbits Earth. That is why we always see the exact same side of the Moon!

Footprints Forever

The Moon has no wind or rain to wash things away. The footprints left by the Apollo astronauts in 1969 are still there today. They will likely last for millions of years!

Make Moon Craters

Fill a shallow tray with flour (about 5 cm deep) and sprinkle a thin layer of cocoa powder on top. Drop small rocks or marbles from different heights into the tray. Observe the craters they make!

Observe: How does the size of the crater change when you drop from higher up? What happens when you drop two rocks close together? Real craters on the Moon were made by asteroids and meteoroids hitting the surface billions of years ago.

1.5

Our Solar System Family

Our Sun has a family of eight major planets orbiting around it. Together with their moons, dwarf planets, asteroids, and comets, they make up the solar system.

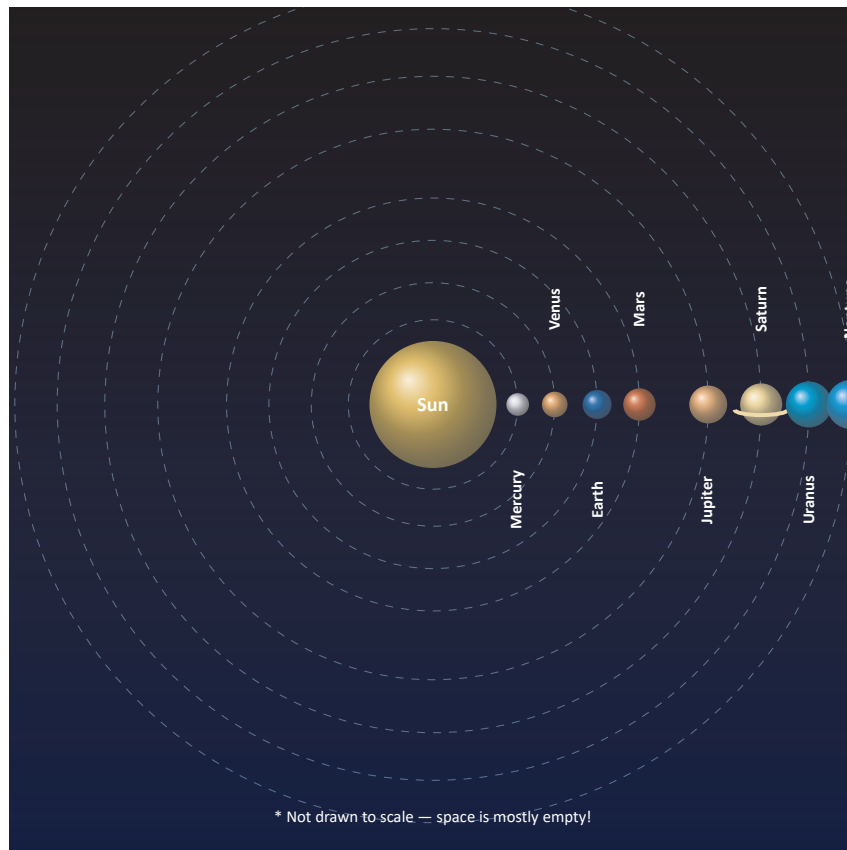


Figure 1.5: Eight planets orbit the Sun at vastly different distances.

The Sun Is Everything

The Sun makes up 99.8% of all the mass in the entire solar system. All the planets, moons, asteroids, and comets together make up only 0.2%!

1.6

The Night Sky

On a clear night, away from city lights, you can see thousands of stars. They look like tiny pinpricks of light, but each one is a massive ball of hot gas, just like our Sun. They only look small because they are unimaginably far away.

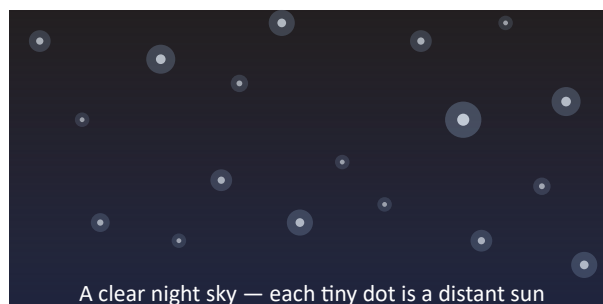


Figure 1.6: Every star you see is a sun, just like ours, shining across deep space.

Stars Twinkle

Stars appear to twinkle because their light is bent as it passes through Earth's moving, windy atmosphere. In outer space, stars do not twinkle at all!

Draw the Sky Tonight

Go outside after sunset with a grown-up. Look up and draw what you see. Include the Moon (what shape is it?), any bright stars, and anything else you notice. Write the date and time on your drawing. Try it again in exactly one week. Has anything changed?

1.7

Chapter Review

Key Ideas

- **Space** begins about 100 km above Earth. It is vast, mostly empty, and contains everything in the universe.
- **Earth** is a planet. It orbits the Sun and is the only known home for life.
- The **Sun** is a star — a giant ball of hot gas that gives us light and heat.
- The **Moon** orbits Earth. It shines by reflecting sunlight.
- Our **solar system** has the Sun, eight planets, and many smaller objects.
- **Stars** are distant suns. They look tiny because they are so far away.

Review Questions

1. Where does space begin?
2. What makes the Moon shine at night?
3. Why is the Sun important for life on Earth?
4. How many planets are in our solar system?
5. Why do stars look so small in the sky?
6. Can you name the planet we live on?

Words to Know

Atmosphere The layer of air and gases around Earth.

Planet A large, round world that orbits a star.

Star A huge ball of hot gas that makes its own light and heat.

Orbit The path one object takes as it moves around another object in space.

Solar System The Sun and everything that orbits around it.

Universe Everything that exists — all space, matter, and energy.

Things to do!

Observe

Go outside tonight and draw the sky. Look for the Moon, bright stars, and any planets. Note the date and time. Come back in a week and compare — what changed? (See the *Draw the Sky Tonight* activity on the previous page for details.)

Think

We see the same stars night after night, yet the universe is constantly moving. If the Earth is spinning at 1,670 km/h and hurtling around the Sun at 107,000 km/h, why don't the star patterns appear to shift wildly every night? What keeps them looking fixed?

Try It

Make a shadow clock: Push a stick into the ground outside. Every hour, mark where the shadow falls and write the time. Watch how the shadow moves, shrinks toward noon, and lengthens again. You have just built a working sundial — one of humanity's oldest scientific instruments!

Explore Further

- **Stellarium (Free Planetarium):** <https://stellarium.org> — See exactly what the sky looks like from your location.
- **NASA Space Place:** <https://spaceplace.nasa.gov> — Games, facts, and activities.
- **ESA Kids:** <https://www.esa.int/kids> — European Space Agency's kid-friendly corner.
- **Spot the ISS:** <https://spotthestation.nasa.gov> — Find out when the space station flies overhead.

★ Did You Know?

Light from the Sun takes only 8 minutes to reach Earth. That means when you feel sunshine on your face, you are feeling sunlight that left the Sun 8 minutes ago! If the Sun suddenly vanished, we would not know about it for 8 whole minutes.

 Challenge

Look at the Moon tonight. Now imagine Earth were the size of a basketball. At that same scale, how far away would the Moon be? Would it fit inside this room? Would you need to walk to the next street? Try to guess before looking up the real numbers!

Questions to Ponder

Think Deeper

1. If you could build a spaceship and visit any planet, which one would you choose and why? What would you pack for the trip?
2. Imagine you met an alien who has never heard of Earth. How would you describe our planet to them in just three sentences?
3. The Sun is 150 million kilometres away. If sunlight takes 8 minutes to reach us, how long would it take a radio message (which also travels at the speed of light) to reach a spacecraft on Mars?
4. If you were standing on the Moon looking up at Earth, what do you think you would see? Would Earth rise and set like the Moon does for us?
5. Why do you think humans have not returned to the Moon since 1972? What would you want to explore there if you were an astronaut?

What You Learned

Things this chapter taught you

- **Space** is the vast, mostly empty area above Earth's atmosphere that contains everything in the universe.
- **Earth** is a rocky planet orbiting the Sun — our cosmic home and the only known place with life.
- The **Sun** is a star that gives us light, heat, and energy through nuclear fusion.
- The **Moon** orbits Earth and shines by reflecting sunlight.
- Our **solar system** includes the Sun, eight planets, moons, asteroids, comets, and dwarf planets.
- **Stars** are distant suns that look tiny only because of their enormous distances from us.



IMAGINE a huge family in space. At the centre is the Sun, a giant star. Orbiting around it are planets, dwarf planets, asteroids, and comets. Together, they make up our solar system — our cosmic home.

What objects live in our solar system, and how are they different?

2.1

The Eight Planets

There are eight major planets in our solar system. In order from the Sun, they are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.

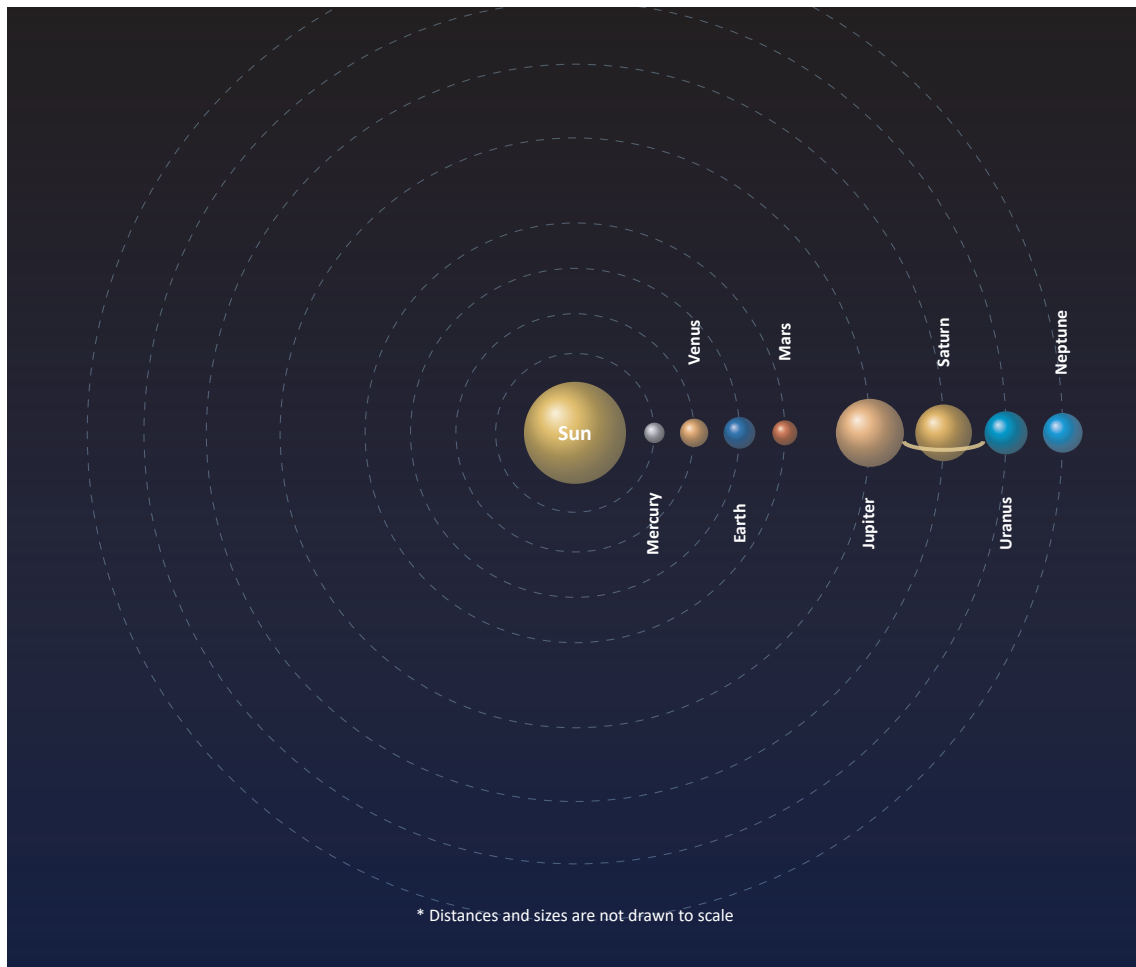


Figure 2.1: The eight planets orbit the Sun at vastly different distances.

Planet

A planet is a large round world that orbits a star. Planets do not make their own light — they shine by reflecting their star's light.

Two Types of Planets

The planets come in two distinct families:

Rocky (Inner) Planets

Mercury, Venus, Earth, Mars

- Close to the Sun
- Small, dense, and rocky
- Solid surfaces
- Few or no moons

Gas & Ice Giants (Outer)

Jupiter, Saturn, Uranus, Neptune

- Far from the Sun
- Large and gaseous/icy
- No solid surface
- Many moons and ring systems

Planet Facts

Planet	Diameter	Day Length	Year Length	Fun Fact
Mercury	4,879 km	59 Earth days	88 days	Smallest planet
Venus	12,104 km	243 Earth days	225 days	Hottest planet
Earth	12,742 km	24 hours	365.25 days	Our home!
Mars	6,779 km	24.6 hours	687 days	Known as the Red Planet
Jupiter	139,820 km	9.9 hours	11.9 years	Biggest planet
Saturn	116,460 km	10.7 hours	29.5 years	Has amazing, visible rings
Uranus	50,724 km	17.2 hours	84 years	Spins entirely on its side
Neptune	49,244 km	16.2 hours	165 years	Has the fastest winds

Mercury to Neptune

A fun way to remember the order of the planets: **My Very Excellent Mother Just Served Us Noodles!**

2.2

A Closer Look: The Inner Planets

Mercury

The smallest planet and closest to the Sun. It is covered in craters and has almost no atmosphere. Days are scorching hot (430°C) and nights are freezing (−180°C).

Venus

The hottest planet! Thick clouds of acid trap heat, making its surface 465°C. Venus spins backwards compared to most planets. A day on Venus is actually longer than its year!

Earth

Our home! The only planet known to harbour life. Liquid water covers 71% of its surface. Earth's perfect distance from the Sun makes it a "Goldilocks" planet — not too hot, and not too cold.

Mars

The famous red planet. It features the largest volcano in the entire solar system — Olympus Mons, which is three times taller than Mount Everest. Mars even has ice caps at its poles, just like Earth!

2.3

A Closer Look: The Outer Planets

● Jupiter

The undisputed king of the planets. Its Great Red Spot is a raging storm bigger than Earth that has been blowing for hundreds of years. Jupiter has 95 known moons!

● Saturn

Famous for its beautiful, wide rings made of ice and rock. Saturn is so incredibly light and un-dense that it would float in a giant bathtub of water! It has 146 moons, including Titan, which has its own thick atmosphere.

● Uranus

An ice giant that spins on its side — its axis is tilted almost 98 degrees! Uranus has 27 known moons, all uniquely named after characters from William Shakespeare's plays.

● Neptune

The windiest, wildest planet. Winds on Neptune can reach 2,100 kilometres per hour — faster than any storm ever recorded on Earth. Neptune is the most distant major planet in our solar system.

2.4

Dwarf Planets

Not everything that orbits the Sun is a planet. Some round objects are simply too small. Astronomers call them **dwarf planets**. The most famous of these is Pluto.

Dwarf Planet

A dwarf planet orbits the Sun and is heavy enough to be round, but it has not “cleared its neighbourhood” (meaning it shares its orbit with lots of other space rocks).

Poor Pluto!

Pluto was considered the ninth planet from its discovery in 1930 until 2006, when astronomers formally decided it should be reclassified as a dwarf planet. It is actually smaller than Earth's Moon!

There are five officially recognised dwarf planets: **Pluto** (the largest), **Ceres** (located in the asteroid belt), **Eris**, **Makemake**, and **Haumea**. Astronomers suspect there may be hundreds more waiting to be found in deep space!

2.5

Asteroids

Asteroids are rocky, airless objects that orbit the Sun. They are much smaller than planets. Most are found clustered together in the **asteroid belt** located between Mars and Jupiter.

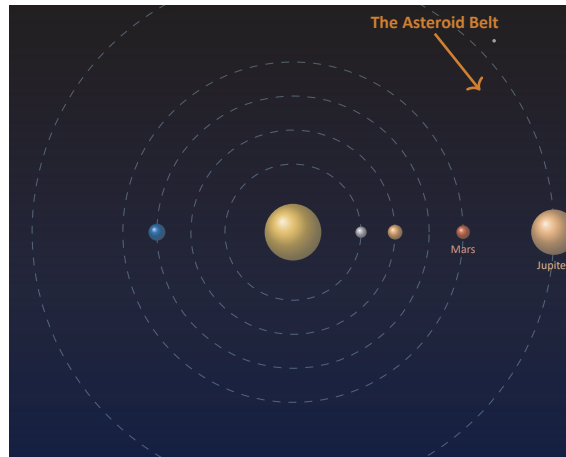


Figure 2.2: The asteroid belt contains millions of rocky leftovers between Mars and Jupiter.

Leftover Building Blocks

Asteroids are basically cosmic leftovers from when the solar system first formed 4.6 billion years ago. They are pieces of rock that never managed to form into a full planet because Jupiter's massive gravity kept pulling them apart!

2.6

Comets — Dirty Snowballs

Comets are chunks of ancient ice, dust, and rock that orbit the Sun in long, stretched-out (elliptical) paths. When a comet gets close to the Sun, the ice heats up, turns into gas, and creates a massive, beautiful glowing tail.

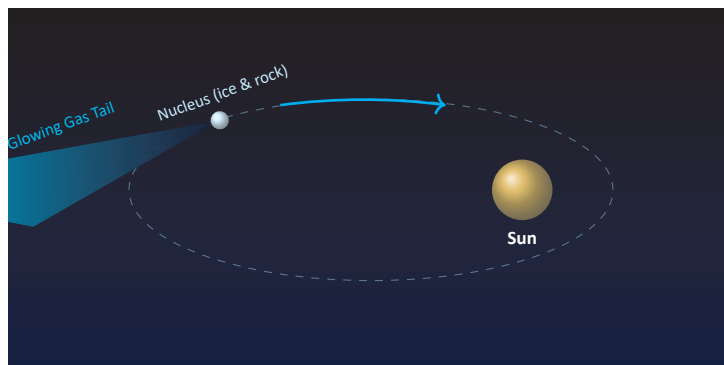


Figure 2.3: Comets have stretched-out orbits. Their tails always point away from the Sun!

Halley's Comet

The most famous comet in history is Halley's Comet. It visits the inner solar system once every 75 to 76 years. It was last seen in 1986 and will return to our skies in 2061. Mark your calendar!

Make a Comet in Your Kitchen

You can create your own comet model! Mix 2 cups of crushed ice, 2 cups of water, 2 tablespoons of dark corn syrup (for organic compounds), 1 tablespoon of vinegar (for gases), and a handful of dirt or sand (for dust). Stir everything together in a bowl. Form it into a lump and place it in the freezer for a few hours.

Observe: Notice how your comet smells (the vinegar represents gases found in real comets) and feels. When you hold it under warm running water, watch how material streams off — just like a real comet's tail near the Sun!

Observe Jupiter's Moons

If you have binoculars or a small telescope, try finding Jupiter in the night sky. It is usually one of the brightest objects after sunset. Even simple binoculars can reveal 3–4 tiny dots lined up next to it — these are its largest moons: Io, Europa, Ganymede, and Callisto! Draw their positions each clear night and notice how they move.

Guide: Use a free app like Stellarium to find Jupiter's exact location on any given night. The moons change position every few hours!

Make a Model Solar System

Create a model of the solar system using everyday items from your home. Use a large ball or balloon for the Sun. Use smaller balls, beads, marbles, or play-dough for the planets. Arrange them in the correct order. Label each planet with its name and one interesting fact.

Challenge: Can you get the distances somewhat accurate? Remember that the outer gas giant planets are spaced *much* further apart from each other than the rocky inner planets are!

2.7

Chapter Review

Key Ideas

- The **solar system** includes the Sun, eight major planets, dwarf planets, asteroids, and comets.

- The four **inner planets** are small and rocky. The four **outer planets** are massive and gaseous/icy.
- **Dwarf planets** like Pluto are round worlds that orbit the Sun but are too small to be true planets.
- **Asteroids** are rocky leftovers found mostly in the asteroid belt.
- **Comets** are icy bodies (“dirty snowballs”) that grow spectacular tails when they approach the Sun’s heat.

Review Questions

1. Name the eight planets in order starting from the Sun.
2. What is the main difference between inner and outer planets?
3. Why was Pluto reclassified as a dwarf planet?
4. Where is the asteroid belt located?
5. Why do comets have tails?
6. Which planet is the biggest, and which is the smallest?

Words to Know

Asteroid A small, rocky object orbiting the Sun.

Comet An icy object that releases gas and grows a tail when near the Sun.

Dwarf Planet A round object orbiting the Sun that is not dominant enough to clear its orbital path.

Gas Giant A massive planet made mostly of gases like hydrogen and helium (Jupiter, Saturn).

Ice Giant A massive planet made mostly of heavier elements, ice, and rock (Uranus, Neptune).

Year The time it takes a planet to complete one full orbit around the Sun.

Things to do!



Observe

Find Jupiter tonight! Look west after sunset for the brightest star-like object. With binoculars, you might see 3–4 tiny dots in a row next to it — those are its largest moons: Io, Europa, Ganymede, and Callisto. Draw their positions each night and watch them dance!

 Think

Jupiter's Great Red Spot is a storm larger than Earth that has been raging for at least 400 years. On Earth, hurricanes last a few weeks at most. What is different about Jupiter that allows storms to last centuries? What would it take for a storm on Earth to last that long?

 Try It

Make a comet in your kitchen! Mix crushed ice, water, dark corn syrup, vinegar, and dirt. Freeze it, then run warm water over it and watch material stream off — just like a real comet's tail near the Sun. (Full recipe in the *Make a Comet in Your Kitchen* activity above.)

 Explore Further

- **NASA Solar System Exploration:** <https://solarsystem.nasa.gov>
- **ESA Kids: Our Solar System:** <https://www.esa.int/kids>
- **Make a comet in a cup:** <https://spaceplace.nasa.gov/comet>
- **Scale model of the solar system:** joshworth.com/pixelspace

 Did You Know?

Saturn is so light that it would actually float in water! Its average density is only 687 kg/m^3 , lower than water's $1,000 \text{ kg/m}^3$. You would need a bathtub big enough to hold it, though — Saturn is 116,460 km wide!

 Challenge

The Sun makes up 99.8% of all the mass in the solar system. All the planets, moons, asteroids, and comets together make up only 0.2%. If you collected every piece of rock and gas in the entire solar system (except the Sun itself), how many Earths worth of material would you have? (Earth's mass is about 0.0003% of the Sun's mass.)

Questions to Ponder

Think Deeper

1. Venus spins backwards (retrograde rotation) compared to most other planets. What do you think could have caused this? Does this change how a day feels on Venus?
2. Jupiter's Great Red Spot is a storm bigger than Earth that has been raging for hundreds of years. Why do you think it has lasted so long? Do you think there are storms on other planets too?
3. If you could redesign Pluto's classification, would you call it a planet or a dwarf planet? What rules would you use to decide what counts as a planet?
4. Saturn is so light it would float in water. What does this tell you about what Saturn is made of? Could you build a spacecraft that lands on Saturn?
5. Comets spend most of their time in the deep, cold outer solar system but briefly visit the warm inner regions. If you were a comet, would you prefer to stay far from the Sun or make the journey inward? Why?

What You Learned

What You Learned

- The **solar system** contains the Sun, eight planets, dwarf planets, asteroids, and comets.
- The four **inner planets** (Mercury, Venus, Earth, Mars) are small and rocky; the four **outer planets** (Jupiter, Saturn, Uranus, Neptune) are massive gas and ice giants.
- **Dwarf planets** like Pluto are round worlds that orbit the Sun but have not cleared their orbital neighbourhood.
- **Asteroids** are rocky leftovers in the asteroid belt between Mars and Jupiter.
- **Comets** are icy bodies that grow glowing tails when they approach the Sun.
- A **year** on any planet is the time it takes to complete one orbit around the Sun.

★ ★ ★

PART II

Everything in the universe is in motion — including us!

EVERY day, the Sun rises and sets. Every night, the Moon changes shape and the stars slowly move across the sky. But nothing is actually moving around us — we are the ones moving! Earth is spinning and travelling through space.

Why does the sky look different every day and night?

3.1

Earth Spins (Rotation)

Imagine standing on a giant ball that slowly spins. That is exactly what Earth does! It rotates around an imaginary line called its **axis**.

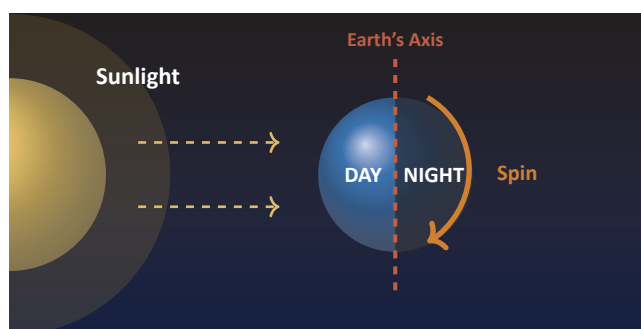


Figure 3.1: Earth spins on its axis. The side facing the Sun experiences day; the side facing away experiences night.

Rotation

Rotation is the spinning of an object around its own axis. One complete rotation of Earth takes 24 hours — which gives us one day.

Faster Than a Jet

Earth spins at about 1,670 kilometres per hour at the equator! But we don't feel it because everything around us — the air, the ground, the buildings — spins smoothly with us.

Why the Sun Rises and Sets

Earth spins from west to east. That is why the Sun appears to rise in the east and set in the west. When you watch the sunrise, you are actually watching your part of Earth turn toward the Sun. When you watch the sunset, you are watching your part of Earth turn away into the shadow!

Different Time, Different Place

When it is daytime in one country, it is nighttime on the opposite side of the world. A child in Australia might be eating breakfast while a child in Brazil is fast asleep!

3.2

Earth Travels (Revolution)

While Earth spins like a top, it also travels around the Sun. This massive journey takes one full year to complete.

Revolution

Revolution is the movement of one object around another. Earth's revolution around the Sun takes about 365.25 days — exactly one year.

Earth moves at an amazing 107,000 kilometres per hour — that is nearly 30 kilometres every single second! Because a year is actually 365 *and a quarter* days long, every four years we add an extra day (29 February) to the calendar to make up for the extra quarter-days. This is called a **leap year**.

3.3

Why We Have Seasons

Why is summer warm and winter cold? Many people mistakenly think it is because Earth is closer to the Sun in summer. **That is not true!** The seasons happen strictly because Earth's axis is tilted.

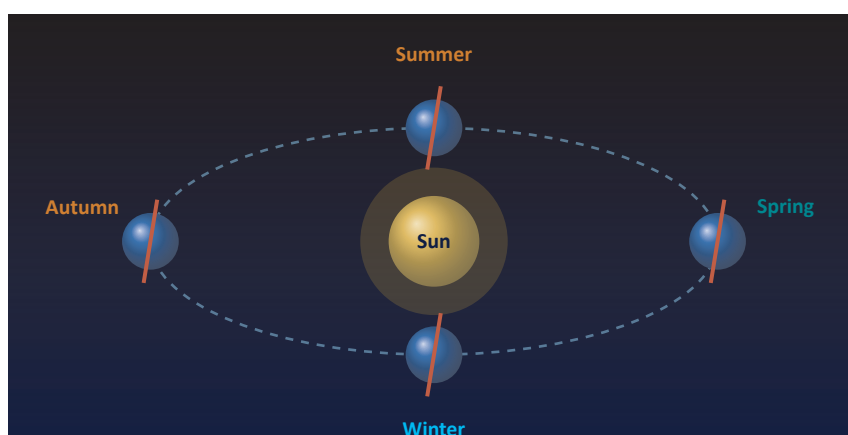


Figure 3.2: Earth's axis is tilted 23.5° . This permanent tilt causes the cycle of summer, autumn, winter, and spring.

Earth's axis is tilted by 23.5 degrees. When the Northern Hemisphere tilts *toward* the Sun, it receives more direct, intense sunlight — resulting in summer. When it tilts *away*, the sunlight hits at a steep angle and is spread out more — resulting in winter.

It Is Not the Distance!

Many people think summer is hot because Earth is physically closer to the Sun. Actually, Earth is closest to the Sun in January (which is winter for the Northern Hemisphere)! The tilt is what determines the season, not the distance.

Make Your Own Seasons

Use a torch (flashlight) and a ball in a dark room. Tilt the ball toward the torch — see how the light hits one part directly in a tight circle? That intense energy creates summer! Tilt it away — now the light spreads out dimly over a wider area. That creates winter!

3.4

Why the Moon Changes Shape

Have you noticed that the Moon does not always look the same? Sometimes it is a thin crescent, sometimes half-lit, and sometimes a full bright circle. These are called **moon phases**.

The Moon orbits Earth about once a month. As it travels around us, we see different amounts of its sunlit side from our viewpoint on Earth.

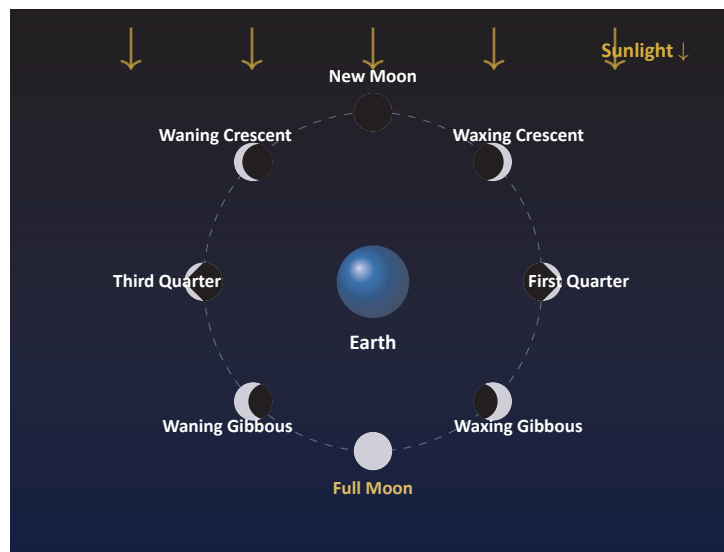


Figure 3.3: The eight phases of the Moon. The outer moons show how the Sun lights them from above.

New Moon The Moon is between Earth and the Sun. Its dark side faces us, so we cannot see it.

Waxing Crescent A thin sliver of light appears on the right side.

First Quarter Half of the Moon is lit — specifically the right half.

Waxing Gibbous More than half lit, and still growing larger.

Full Moon Earth is between the Moon and Sun. The whole face is brightly lit!

Waning Gibbous More than half lit, but now it is shrinking.

Third Quarter The left half of the Moon is lit.

Waning Crescent A thin sliver of light remains on the left side before vanishing.

Waxing vs Waning

A helpful trick for the Northern Hemisphere: when the Moon is **waxing** (growing), the lit side is on the **right**. When it is **waning** (shrinking), the lit side is on the **left**.

Model a Solar Eclipse

You can recreate a solar eclipse at home! Use a torch (flashlight) as the Sun, a small ball (like a tennis ball) as the Moon, and your own head as Earth. In a dark room, hold the ball between the torch and your face. Move it slowly until it blocks the light. That is exactly what happens during a solar eclipse!

Observe: Notice how the ball casts a shadow. Move it closer and farther from your face — how does the shadow change? Real solar eclipses are visible only from a narrow path on Earth because the Moon's shadow is quite small!

Track the Sunrise Position

For one week, wake up early and note where the Sun rises. Pick a fixed landmark (like a tree or building) and mark the sunrise position each morning relative to it. Record your observations.

Observe: Does the Sun rise at exactly the same spot every day? As seasons change, the sunrise point shifts north or south. This happens because of Earth's tilted axis — the same reason we have seasons!

3.5**Eclipses — When Shadows Fall**

Sometimes the Sun, Earth, and Moon line up perfectly in a straight line. When this happens, we experience an eclipse — one of the most stunning sights in nature!

Solar Eclipse

A solar eclipse happens when the **Moon** passes exactly between Earth and the Sun. The Moon blocks the Sun's light, casting a deep shadow on Earth. For a few minutes, day turns into an eerie twilight!

Never Look at a Solar Eclipse!

Even during an eclipse, the Sun is dangerously bright. Never look at a solar eclipse without certified solar eclipse glasses. Regular sunglasses will NOT protect your eyes!

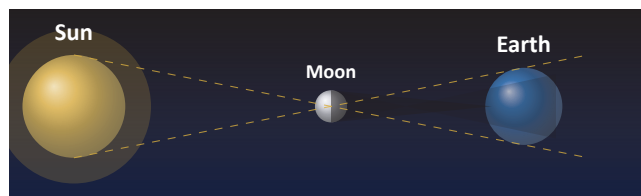


Figure 3.4: During a solar eclipse, the Moon's shadow falls onto Earth, blocking the Sun.

Lunar Eclipse

A lunar eclipse happens when **Earth** passes exactly between the Sun and the Moon. Earth's massive shadow engulfs the Moon, turning it dark.

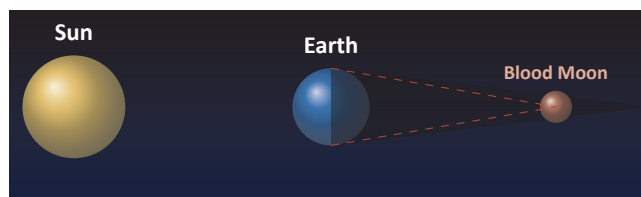


Figure 3.5: During a lunar eclipse, Earth blocks the Sun. Bending light turns the Moon a deep, rusty red.

During a total lunar eclipse, the Moon often turns a deep, eerie red colour. This is because some sunlight bends through Earth's atmosphere and reaches the Moon. The blue light scatters away in our atmosphere, leaving only red light to hit the lunar surface. That is why lunar eclipses are famously called **blood moons**.

Safe to Watch!

Unlike solar eclipses, lunar eclipses are perfectly safe to watch with your bare eyes! They happen about twice a year and can be seen from anywhere on the night-side of Earth.

Observe the Moon for a Month

For one full month, look at the Moon every clear night. Draw its shape in a notebook and record the date and time. At the end of the month, look at your drawings in order. Can you see the pattern? Which phase did you start with, and which did you end with?

3.6

Chapter Review

Key Ideas

- **Rotation** is Earth spinning on its axis — one spin equals one day.
- **Revolution** is Earth orbiting the Sun — one orbit equals one year.
- **Seasons** are caused by Earth's permanent 23.5° axis tilt, not by distance from the Sun.
- **Moon phases** occur because we see different portions of the Moon's sunlit side as it orbits us.
- A **solar eclipse** happens when the Moon blocks the Sun from our view.
- A **lunar eclipse** happens when Earth's shadow falls on the Moon, turning it red.

Review Questions

1. What is the difference between rotation and revolution?
2. What causes the seasons? Why is summer hotter than winter?
3. Draw the eight phases of the Moon in order, starting from a new moon.
4. Why does a solar eclipse happen? Why is it dangerous to look at?
5. Why does the Moon turn red during a lunar eclipse?
6. How fast does Earth travel through space?

Words to Know

Axis An imaginary line running straight through the centre of Earth from pole to pole.

Eclipse An event where one celestial body temporarily blocks light from reaching another.

Leap Year A calendar year with 366 days, happening once every four years.

Lunar Eclipse When Earth's shadow falls on the Moon.

Phase The specific shape of the lit part of the Moon as seen from Earth.

Solar Eclipse When the Moon's shadow falls directly on Earth.

Waning When the lit portion of the Moon is visibly getting smaller.

Waxing When the lit portion of the Moon is visibly getting larger.

Things to do!

Observe

Track the Moon for a month. Every clear night, go outside and look at the Moon. Draw its exact shape in a notebook with the date and time. After a full cycle (about 29 days), look at your drawings in order. You will have mapped all eight phases yourself — just like ancient astronomers did!

Think

The Moon is slowly drifting away from Earth at 3.8 cm per year. Billions of years ago, it was much closer and appeared much larger in the sky. What would it have been like to stand on Earth when the Moon was 10 times closer? How big would solar eclipses have been?

Try It

Model the seasons with a torch and a ball. In a dark room, shine a flashlight at a tilted ball. Notice how the light hits the tilted side directly (summer) and the opposite side at a shallow angle (winter). This single tilt is the reason Earth has seasons — not distance from the Sun!

Explore Further

- **NASA Space Place: Moon Phases:** <https://spaceplace.nasa.gov/moon-phases/>
- **Track Daylight Where You Live:** <https://www.timeanddate.com/sun/>
- **Find Your Shadow's Length:** <https://spaceplace.nasa.gov/shadows/>
- **Watch Solar Eclipse Videos:** <https://www.nasa.gov/eclipse>

 Did You Know?

A day on Venus is longer than a year on Venus! Venus takes 225 Earth days to orbit the Sun, but it takes 243 Earth days to spin once on its axis. That means a single day on Venus lasts longer than its entire year. Even stranger: Venus spins backwards, so the Sun rises in the west and sets in the east!

 Challenge

If Earth stopped spinning suddenly, everything on the surface would keep moving east at 1,670 km/h — like passengers in a crashing car. But there is a more subtle effect: Earth's rotation causes it to bulge at the equator. If it stopped, the oceans would flow toward the poles. Which continents would gain land, and which would be submerged?

Questions to Ponder

Think Deeper

1. Earth spins at 1,670 km/h at the equator. If Earth suddenly stopped spinning, what do you think would happen to everything on the surface? Would we feel it?
2. The Moon is slowly moving away from Earth — about 3.8 centimetres every year. What will happen to the length of a day millions of years from now? Will solar eclipses still be possible?
3. If Earth had no tilt at all (0 degrees instead of 23.5 degrees), what would the seasons be like? Would any place on Earth ever experience winter?
4. During a total solar eclipse, birds sometimes stop singing and the temperature drops noticeably. Why do you think animals react this way? What would you do if you experienced a sudden midday darkness?
5. People in the Southern Hemisphere see the Moon upside down compared to people in the Northern Hemisphere. If you video-called a friend in Australia and pointed at the Moon, what would they see?

What You Learned

Things this chapter taught you

- **Rotation** is Earth spinning on its **axis** — one full spin gives us a 24-hour day.
- **Revolution** is Earth orbiting the Sun — one full orbit gives us a 365.25-day year (with **leap years** every four years).
- **Seasons** are caused by Earth's 23.5° axis tilt, not by how far we are from the Sun.
- **Moon phases** (waxing and waning) happen as we see different portions of the Moon's sunlit side.
- A **solar eclipse** occurs when the Moon blocks the Sun; a **lunar eclipse** occurs when Earth's shadow falls on the Moon.



WHY does the Moon orbit Earth? Why do we stay on the ground instead of floating away? The answer is **gravity** — an invisible force that pulls everything toward everything else.

How does gravity work, and how do we travel in space?

4.1

What Is Gravity?

Gravity is the force that pulls objects together. The more mass an object has, the stronger its gravity. The Sun has the strongest gravity in the entire solar system, which is why all the planets continuously orbit around it.

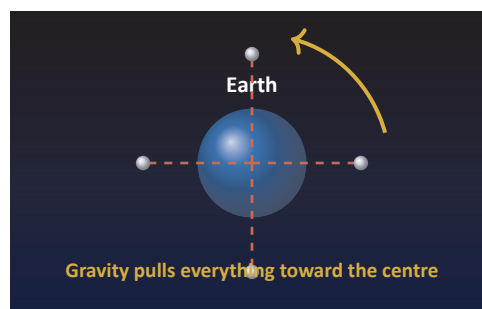


Figure 4.1: Gravity pulls all matter toward the centre of Earth.

Gravity

Gravity is an invisible force that pulls objects with mass toward each other. The more mass an object has, the stronger its gravity. Earth's gravity keeps us safely on the ground and keeps the Moon locked in its monthly orbit.

Weight vs Mass

Your **mass** (how much actual matter is in your body) stays exactly the same everywhere in the universe. Your **weight** (how strongly gravity pulls you down) changes! On the Moon,

you would weigh only one-sixth of your Earth weight because the Moon is smaller and has weaker gravity.

Gravity in the Solar System

The Sun's immense gravity holds the entire solar system together. Each individual planet's gravity, in turn, keeps its own moons in orbit. That is why everything stays gracefully in its place!

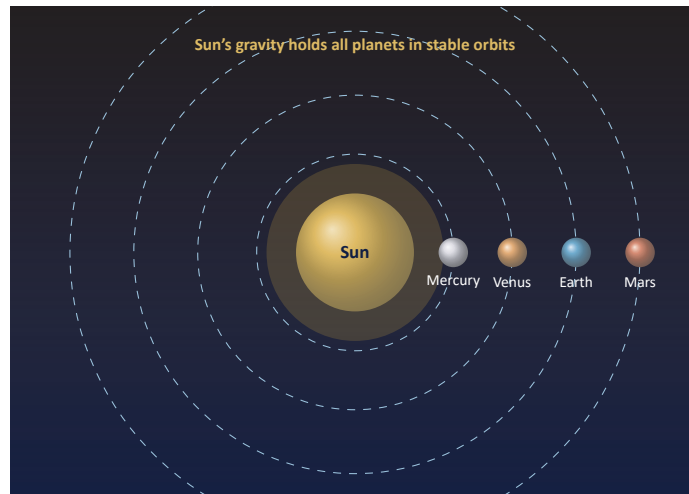


Figure 4.2: The Sun's powerful gravity keeps all the planets traveling in predictable orbits.

4.2

Orbits — Falling Around

An orbit happens when an object moves sideways so fast that it constantly misses the ground as it falls! Think of throwing a baseball — it curves down toward the earth. Now imagine throwing it so incredibly hard that the downward curve matches the curve of Earth itself. The ball would keep falling around Earth forever without ever landing. That is what an orbit is!

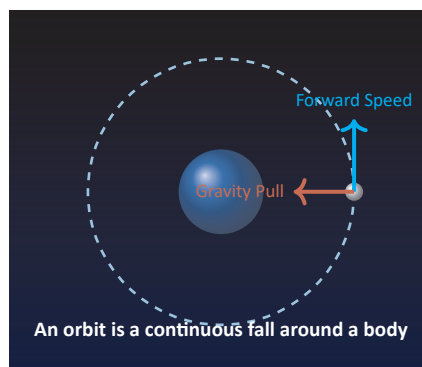


Figure 4.3: Spacecraft in orbit constantly fall toward Earth but move sideways fast enough to keep missing it. The forward speed and gravity together create a curved path.

International Space Station

The ISS orbits Earth completely once every 90 minutes! Astronauts on board experience weightlessness not because there is zero gravity up there, but because they are in a state of free fall, constantly falling around Earth together with the station.

4.3**How Rockets Work**

Getting into space is immensely difficult because you have to fight Earth's gravity. That is why we use rockets — the most powerful engineering machines ever built!

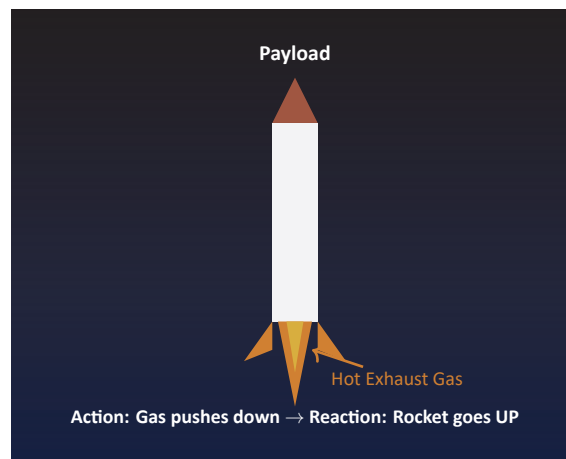


Figure 4.4: A rocket uses Newton's Third Law, shooting gas downward to thrust upward.

Rockets work by **action and reaction** (Newton's Third Law). The engine burns liquid fuel to create high-pressure hot gases that blast out of the nozzle at tremendous speeds. The action is the gas shooting downward; the reaction is the rocket being thrust upward. This works even in the vacuum of space because the rocket carries its own supply of oxygen!

Saturn V

The legendary Saturn V rocket that launched astronauts to the Moon was as tall as a 36-storey building (111 metres)! Its five powerful engines burned 15 tonnes of fuel every single second.

Stages

Most large rockets are built in **stages**: massive fuel tanks and engines that drop off into the ocean once they are empty. This sheds dead weight, allowing the remaining rocket to go faster and reach orbit.

4.4

Exploring the Solar System

Humans have sent robotic spacecraft and probes to every single planet in our solar system. Here are some of history's most famous missions:

Apollo 11 — 1969

The first mission to successfully land humans on the Moon. Astronauts Neil Armstrong and Buzz Aldrin walked on the dusty lunar surface while command pilot Michael Collins orbited above. "One small step for a man, one giant leap for mankind."

Voyager — 1977

Two twin spacecraft sent to explore Jupiter, Saturn, Uranus, and Neptune. Voyager 1 has now crossed the edge of our solar system and is traveling through interstellar space! Each carries a golden phonograph record containing sounds, music, and images of Earth.

Mars Rovers

Since 1997, NASA has deployed robotic rovers across the Martian landscape. Opportunity lasted an incredible 14 years, while Perseverance landed in 2021 carrying Ingenuity — the first powered aircraft to fly on another planet.

Extreme Speeds

The fastest human-made object in history, the Parker Solar Probe, travels at an astonishing 690,000 kilometres per hour as it swoops close to the Sun!

Drop and Feel Gravity

Hold two different objects — a heavy book and a pencil — at the same height. Drop them at exactly the same time. Which one hits the ground first? Try it with a crumpled piece of paper and a flat sheet of paper. What changes?

Observe: Without air resistance, all objects fall at the same rate! This is why astronauts on the Moon dropped a hammer and a feather together — they landed at the exact same time. Try rolling a ball off a table while dropping another ball straight down from the same height. Which hits the ground first?

Build a Straw Rocket

Make a simple rocket from a drinking straw and a strip of paper. Cut a paper rectangle (about 4 cm × 6 cm), wrap it loosely around the straw to form a cylinder, and tape it closed. Fold one end into a cone and tape it shut. Add paper fins to the other end. Blow into the straw to launch your rocket!

Experiment: Try changing the fin size, nose cone shape, or rocket length. Which design

flies farthest? This is exactly what real rocket engineers do — test different shapes to make more efficient rockets!

4.5

Life in Space

What is it like to live aboard a space station? Astronauts on the ISS experience **microgravity** (near weightlessness). Everything floats freely — including food packets, tools, and the astronauts themselves! They must strap themselves down into sleeping bags at night, and they exercise for two hours every day to keep their bones and muscles strong.

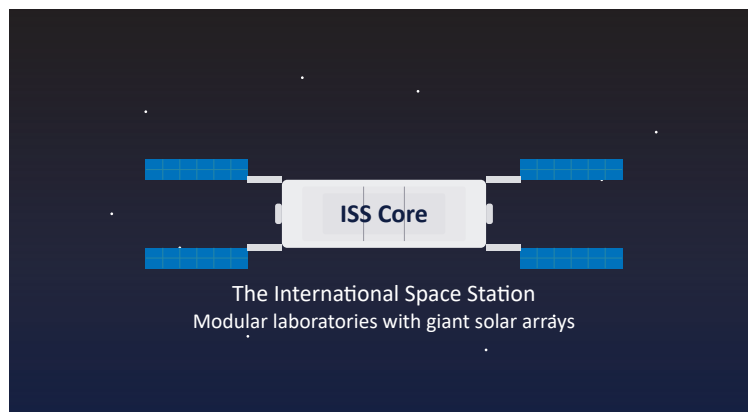


Figure 4.5: The ISS features modular laboratories and giant solar panels to generate power.

Space Debris

There are thousands of pieces of defunct satellites, spent rocket boosters, and tiny paint chips orbiting Earth at bullet speeds. This is known as **space debris**. Even a tiny speck can punch a hole through a spacecraft window because it moves so fast. Scientists are actively developing innovative ways to clean up space junk!

Find the ISS Online

Visit <https://spotthestation.nasa.gov> and enter your hometown. Find out when the ISS will pass directly over your location tonight. Go outside at the appointed time and look for a bright, steady light gliding silently across the stars!

4.6

Chapter Review

Key Ideas

- **Gravity** is an invisible force pulling masses together; more mass creates stronger gravity.
- An **orbit** is achieved when an object moves sideways fast enough to continuously miss the ground as it falls.
- **Rockets** operate via action and reaction: hot exhaust gas shoots down, pushing the rocket upward.
- **Spacecraft** have explored every planet in our solar system, with Voyager probing interstellar space.
- Astronauts live and work on the **ISS** in a microgravity environment.

Review Questions

1. What is gravity, and what property of an object does its strength depend on?
2. Why do astronauts float inside the International Space Station?
3. How do rockets work according to Newton's Third Law?
4. Which historic space mission first landed humans on the Moon?
5. What is space debris, and why is it dangerous to active satellites?
6. Would your mass or your weight change if you visited the Moon? Why?

Words to Know

Action and Reaction Newton's Third Law: for every action, there is an equal and opposite reaction.

Gravity The invisible attractive force between all objects that possess mass.

ISS International Space Station — a habitable scientific laboratory orbiting Earth.

Microgravity A condition where gravitational pull is very weak, making objects appear weightless.

Orbit The curved trajectory of a celestial body or spacecraft around another body.

Rocket A vehicle propelled by the exhaust of high-pressure rocket engines.

Space Debris Abandoned human-made hardware and fragments orbiting Earth.

Things to do!

Observe

Spot the ISS! Visit <https://spotthestation.nasa.gov>, enter your city, and find out when the space station will fly over tonight. Go outside at the right time and look for a bright, steady light moving silently across the sky — faster than any airplane, with no blinking lights. That is a 100 billion laboratory flying with people inside!

Think

Astronauts on the ISS float not because there is no gravity, but because they are in free fall. The ISS is falling toward Earth constantly — but moving sideways so fast that it keeps missing the ground. If you were in a falling elevator, would you float too? What is the difference between falling 100 metres and falling all the way around Earth?

Try It

Drop two objects of different weights from the same height. A book and a pencil, for example. Which hits the ground first? (Try it with a flat sheet of paper vs. a crumpled ball too.) Without air resistance, all objects fall at the same rate — on the Moon, an astronaut dropped a hammer and a feather, and they landed together!

Explore Further

- **NASA Space Place: Gravity:** <https://spaceplace.nasa.gov/gravity/>
- **NASA Rocket Simulator:** <https://www.grc.nasa.gov/www/k-12/rocket/>
- **Spot the ISS:** <https://spotthestation.nasa.gov>
- **ESA Kids Space:** <https://www.esa.int/kids>
- **Watch Live Launches:** <https://www.nasa.gov/nasatv>

★ Did You Know?

There are over 27,000 pieces of space debris larger than a softball currently being tracked by space agencies. They travel at speeds of up to 28,000 km/h — fast enough for a fleck of paint to crack a spacecraft window! To avoid collisions, the ISS sometimes fires its engines to dodge incoming junk.

Challenge

The Voyager 1 spacecraft, launched in 1977, is now over 24 billion km away and still traveling. It carries a Golden Record with sounds and images of Earth — intended for any alien civilization that finds it. At its current speed of 61,000 km/h, how long will it take to reach the nearest star (Proxima Centauri, 40 trillion km away)? Would any human alive today still be alive when it arrives?

Questions to Ponder

Think Deeper

1. Astronauts on the ISS float not because there is no gravity, but because they are in free fall around Earth. If you were in a falling elevator, would you float too? What is the difference between falling and orbiting?
2. We have sent robots to Mars, Venus, and beyond, but humans have only been to the Moon (max 400,000 km away). Why is sending humans so much harder? What would we need to build to safely send people to Mars?
3. Imagine you are designing a space mission to Jupiter's moon Europa, which has a liquid ocean under its icy crust. What kind of robot would you build to explore it? How would it get through the ice?
4. Space debris travels at about 28,000 km/h. At that speed, even a fleck of paint can crack a spacecraft window. If you were in charge of cleaning up space junk, what solution would you propose?
5. If you could send one message to extraterrestrial life (like the Voyager Golden Record), what would you include? What sounds, images, or ideas would best represent life on Earth?

What You Learned

Things this chapter taught you

- **Gravity** is an invisible force that pulls all objects with mass toward each other.
- Your **mass** stays the same everywhere; your **weight** changes depending on the gravity of the world you are on.
- An **orbit** is a continuous free fall — moving sideways so fast that you keep missing the ground.

- **Rockets** work by **action and reaction** (Newton's Third Law): exhaust shoots down, pushing the rocket up.
- Astronauts on the **ISS** experience **microgravity** because they are in free fall around Earth.
- **Space debris** is human-made junk orbiting Earth and poses a danger to satellites and spacecraft.



PART III

Look up at the night sky and journey to the farthest reaches of the universe.

ON a clear, dark night, look up into the sky. Hundreds of tiny, brilliant lights sparkle against the blackness of space. Each one of those glittering points is a sun—a colossal sphere of incandescent gas just like our own Sun, located at an almost unimaginable distance from Earth. Let us embark on a journey to explore the stars.

What are stars, and how can we use them to find our way around the night sky?

5.1
What Is a Star?

At its core, a star is a massive ball of glowing, superheated gas that generates its own light and heat. Deep inside a star’s core, gravity squeezes hydrogen atoms together with such immense force that they fuse to form helium. This process is called **nuclear fusion**, and it releases an extraordinary amount of energy. That energy radiates outward through the vacuum of space as the light we see and the warmth we feel.

Star

A star is a giant, luminous sphere of plasma (superheated gas) that produces its own light and heat through the process of nuclear fusion. Our Sun is a medium-sized star.

Star Colours and Temperatures

If you look closely at the night sky, you might notice that stars do not all shine with the same white light—they come in a spectrum of colours! A star’s colour is actually a cosmic thermometer telling you its surface temperature.



Figure 5.1: The classification of star colours, ranging from the hottest blue stars to the coolest red dwarfs.

Our Sun is a yellow star, sitting right in the middle of the temperature scale at roughly $5,500^{\circ}\text{C}$. Blue supergiants burn furiously at temperatures exceeding $30,000^{\circ}\text{C}$, while gentle red dwarfs smoulder at around $3,000^{\circ}\text{C}$.

The Nearest Star

The closest star system to Earth (after our Sun) is Alpha Centauri, and its nearest member is a small red dwarf called Proxima Centauri, located 4.2 light-years away. This means light from Proxima Centauri traveling at $300,000\text{ km/s}$ takes over four years to reach our eyes. If you boarded the fastest spacecraft ever built by humans, the journey would still take roughly 70,000 years!

The Scale of Stars

Stars span a breathtaking range of sizes. The smallest active stars, known as red dwarfs, are only slightly larger than the planet Jupiter. At the other extreme exist red supergiants like Betelgeuse—if you placed one at the centre of our solar system, it would engulf Mercury, Venus, Earth, and extend past the orbit of Mars!

How Many Stars Are Out There?

Standing under a dark, cloudless sky far from city lights, your unaided eyes can distinguish roughly 2,500 individual stars. However, that is just a tiny fraction of our local cosmic neighbourhood. Across the observable universe, astronomers estimate there are more stars than there are grains of sand on every beach and desert on Earth combined!

5.2

Constellations — Pictures in the Sky

For thousands of years, stargazers across the globe have played a game of cosmic connect-the-dots, linking stars together to outline gods, beasts, and heroes. We call these star patterns **constellations**. Because different civilizations projected their own myths onto the heavens, ancient Greek astronomers visualized a mighty hunter named Orion where traditional Chinese astronomers saw the patterns of the Three Stars and the Jade Well.

Today, astronomers divide the celestial sphere into 88 official constellations, creating a permanent map to help navigate the night sky.

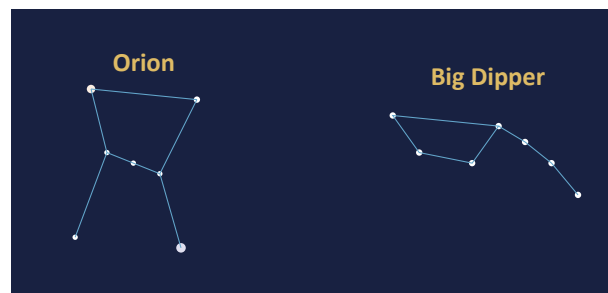


Figure 5.2: Two of the most recognizable star patterns: Orion the Hunter (left) and the Big Dipper (right).

Finding North with the Big Dipper

The Big Dipper is more than just a famous shape—it is a celestial compass! The two outer stars forming the front of the dipper’s bowl (Merak and Dubhe) are called “pointer stars.” If you draw an imaginary line through them and extend it upward, it points directly to Polaris, the North Star. Because Polaris sits almost perfectly above Earth’s North Pole, it appears stationary in the sky while the entire star field rotates around it throughout the night.

88 Official Constellations

In 1922, the International Astronomical Union adopted 88 official constellations that cover the entire sky like puzzle pieces. Among the most celebrated are **Ursa Major** (the Great Bear, containing the Big Dipper), **Orion** (the Hunter), **Cassiopeia** (the Vain Queen), **Scorpius** (the Scorpion), and **Cygnus** (the Swan).

The Path of the Zodiac

You have likely heard of the signs of the zodiac—Aries, Taurus, Gemini, Leo, and their companions. In astronomy, these twelve famous constellations form a belt along the **ecliptic**, which is the apparent path the Sun traces across the sky over the course of a year. When someone says their astrological sign is Leo, it simply means that from Earth’s perspective, the Sun was passing in front of the constellation Leo during the month they were born!

5.3

Galaxies — Star Cities

Stars are not scattered randomly through the void of space like dust. Instead, gravity binds them together into vast, sprawling communities called **galaxies**. A single galaxy can house anywhere from a few hundred million to several trillion stars, along with vast clouds of gas, interstellar dust, and mysterious dark matter.

Galaxy

A galaxy is an immense gravitationally bound system of stars, stellar remnants, interstellar gas, dust, and dark matter. Our solar system resides inside a galaxy called the Milky Way.

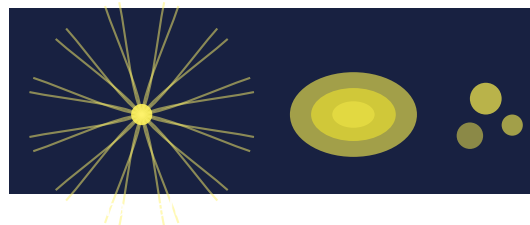


Figure 5.3: The three primary morphological classifications of galaxies: spiral, elliptical, and irregular.

Astronomers classify galaxies into three main shapes:

- **Spiral Galaxies:** Majestic, rotating discs with graceful pinwheel arms winding out from a bright central core.
- **Elliptical Galaxies:** Egg-shaped or spherical swarms of older stars with very little gas or dust to form new stars.
- **Irregular Galaxies:** Chaotically shaped systems that lack a distinct form, often distorted by the gravitational pull of larger neighbouring galaxies.

5.4

Our Galactic Home: The Milky Way

We live inside a barred spiral galaxy known as the **Milky Way**, a spinning stellar metropolis containing between 100 billion and 400 billion stars. Because we reside within the galactic disc, viewing our galaxy from Earth is like standing inside a dense forest—we see the surrounding trees stretching across our field of view as a luminous, milky river of light spilling across the night sky.

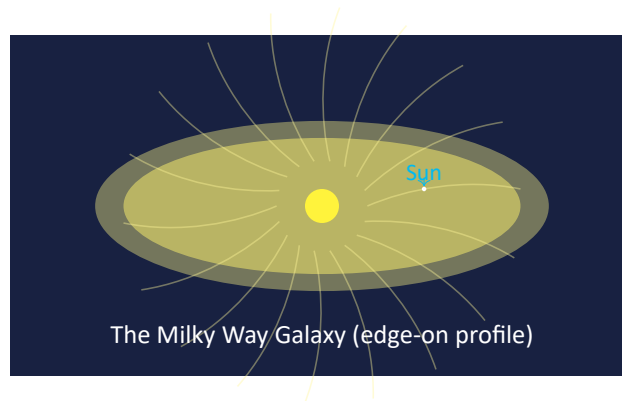


Figure 5.4: An edge-on view of the Milky Way, illustrating its flat, disc-like structure and bulging central core.

Our Sun is situated safely in the galactic suburbs, roughly halfway between the blazing centre and the outer edge, on a structure known as the Orion Spur. Even traveling at a mind-boggling 828,000 km/h, our solar system takes about 230 million years to complete a single orbit around the galactic core—a journey known as one “cosmic year.”

5.5

Tips for Practical Stargazing

You do not need an expensive telescope or advanced instruments to explore the cosmos! The naked eye is one of the greatest astronomical tools available. Here is how to make the most of an evening under the stars:

Find Orion at Night

On a clear winter evening, step outside and face south (if you are in the Northern Hemisphere). Look for three bright stars aligned in a neat, diagonal straight line—this is Orion's Belt. Just above the belt sit two bright stars marking his shoulders (including fiery red Betelgeuse). Below the belt are two bright stars forming his knees (including icy blue Rigel). Can you trace the outline of the celestial hunter?

- **Locate the Big Dipper:** Look toward the northern sky to spot the seven bright stars forming a giant drinking ladle, and use the pointer stars to locate Polaris.
- **Spot the Pleiades:** Scan Taurus for a glittering cluster of young stars often called the "Seven Sisters." It looks like a miniature, misty dipper!
- **Catch a Shooting Star:** Keep your eyes peeled for meteors—tiny specks of space rock burning up in Earth's atmosphere—especially during predictable annual showers like the Perseids or Geminids.
- **Trace the Milky Way:** Venture out on a moonless summer night away from city illumination to witness the hazy, glowing backbone of our home galaxy.
- **Know When to Look:** The best stargazing falls within a few days of the new Moon, when moonlight does not wash out fainter stars. Check a lunar calendar and plan your sessions around moonrise and moonset.
- **Find Planets vs. Stars:** Planets shine with a steady, unwavering light, while stars twinkle due to atmospheric turbulence. Spot Venus blazing low in the west after sunset, or Jupiter riding high as the brightest object after the Moon sets.
- **Scan with Binoculars:** Even ordinary 7×35 or 10×50 binoculars transform the night sky—they reveal dozens more stars in the Pleiades, the four largest moons of Jupiter, the phases of Venus, and even the Andromeda Galaxy as a faint fuzzy ellipse.
- **Spot the ISS:** The International Space Station glides overhead as a bright, silent, steady light—much faster than an aeroplane, with no blinking navigation lights. Sign up for NASA's "Spot the Station" alerts for your location.
- **Dress Warm and Stay Comfortable:** Night air cools rapidly after sunset. Layer your clothing, bring a blanket or reclining chair, and use a red-light flashlight (tape red cellophane over a regular torch) to read star charts without ruining your night vision.

The Challenge of Light Pollution

Artificial illumination from streetlights, buildings, and cars scatters into the atmosphere, creating a synthetic glow known as **light pollution**. This washes out the subtle starlight of the night sky. In a heavily lit metropolis, you might only see a dozen stars, but from a certified Dark Sky Reserve, thousands of stars and the textured dust lanes of the Milky Way burst into view!

Use Your Hand as a Sky Ruler

You can measure distances between stars using only your hand! Hold your arm out at full length:

- Your pinky finger covers about 1 degree of sky.
- Your three middle fingers together cover about 5 degrees.
- Your fist covers about 10 degrees.
- Your spread hand (thumb to pinky) covers about 20 degrees.

Go outside and measure: how many fists wide is the Big Dipper? How many pinky-widths between the two pointer stars of the Big Dipper? The full Moon is only 0.5 degrees across — that is half a pinky-width!

Spot Star Colours with Your Eyes

On a clear, dark night, look at the brightest stars and try to notice their colours. Betelgeuse (in Orion) often appears slightly orange or reddish. Rigel (also in Orion) looks bluish-white. Sirius (the brightest nighttime star) sparkles with blue and white flashes.

Guide: Use a star chart app to identify these stars. The colours are caused by temperature — blue stars are much hotter than red stars. Can you find a yellow star like our Sun?

Create Your Own Star Wheel

Build a rotating planisphere (star finder) to identify which constellations are visible at any specific hour of the year! Download and print a free template from NASA at <https://spaceplace.nasa.gov/starfinder/> and follow the simple assembly instructions. Take it outside with a red-light flashlight (which preserves your night vision) and test your celestial navigation skills!

5.6

Chapter Review

Key Concepts

- **Stars** are massive spheres of superheated gas that produce light and heat through nuclear fusion at their cores. Our Sun is a typical medium-sized star.
- A star's **colour** indicates its surface temperature: blue stars are the hottest, while red dwarfs and supergiants are the coolest.
- **Constellations** are recognizable patterns of stars that astronomers use to map and navigate the celestial sphere. There are 88 official constellations.

- **Galaxies** are sprawling cosmic cities composed of billions of stars, gas, dust, and dark matter bound by gravity. We live in a spiral galaxy called the Milky Way.
- Stargazing requires no special equipment—understanding light pollution and using tools like pointer stars and planispheres can open up the entire night sky to unaided eyes.

Review Questions

1. What atomic process occurs inside a star's core to make it shine?
2. If you observe a bright blue star and a reddish star in the sky, which one has a higher surface temperature?
3. What is the difference between an informal star picture and an official constellation? How many recognized constellations exist today?
4. What is the closest star system to our solar system, and roughly how long does its light take to reach Earth?
5. What are the three primary shapes used to classify galaxies?
6. Explain how artificial light pollution affects our view of the night sky.

Words to Know

Constellation An officially recognized group of stars that forms a distinct pattern and represents a specific region of the celestial sphere.

Ecliptic The apparent annual path that the Sun traces across the sky against the background of stars.

Galaxy A massive, gravitationally bound system containing millions to trillions of stars, gas, dust, and dark matter.

Light Pollution Excessive or obtrusive artificial light generated by human civilization that obscures the natural night sky.

Milky Way The barred spiral galaxy that serves as home to our solar system and over 100 billion other stars.

Nuclear Fusion The high-energy atomic reaction powering stars, wherein hydrogen nuclei combine under immense heat and pressure to create helium and release photons.

Planisphere An adjustable, two-layered star chart used to display the visible stars and constellations for any date and time.

Red Dwarf A small, relatively cool, and faint star; the most abundant type of star in the universe.

Supergiant An exceptionally massive and luminous star nearing the end of its stellar lifecycle.

Things to do!

Observe

Find Orion tonight! On a clear winter evening, look south for three bright stars in a straight, diagonal line — Orion's Belt. Above the belt are two shoulder stars (the reddish one is Betelgeuse). Below the belt are two knee stars (the bluish one is Rigel). Can you trace the hunter's outline? If you have binoculars, look at the fuzzy patch below the belt — that is the Orion Nebula, a stellar nursery!

Think

Different ancient civilizations saw different patterns in the same stars. The Greeks saw Orion the Hunter; traditional Chinese astronomers saw the Three Stars and the Jade Well. If you could invent your own constellation from the stars visible tonight, what shape would you connect? What story would you tell?

Try It

Use your hand as a sky ruler! Hold your arm out straight. Your fist covers about 10 degrees of sky. Your pinky covers about 1 degree (the Moon is only 0.5 degrees!). Go outside and measure: how many fists wide is the Big Dipper? How many pinkies between the two pointer stars? You are doing real angular measurement — the same technique astronomers use!

Explore Further

- **Stellarium Web (Free Online Planetarium):** <https://stellarium-web.org>
- **NASA Space Place — Stars:** <https://spaceplace.nasa.gov/stars/>
- **Hubble Space Telescope — Stellar Imagery:** <https://hubblesite.org>
- **Interactive Light Pollution Map:** <https://www.lightpollutionmap.info>
- **BBC Sky at Night Magazine:** <https://www.skyatnightmagazine.com>

★ Did You Know?

Stars twinkle, but planets do not! Stars are so far away that they are just points of light — Earth's turbulent atmosphere bends that point, making it dance. Planets are close enough to be tiny discs, so the bending averages out. Spot the difference: a twinkling light is a star; a steady, calm light is a planet!

 Challenge

The nearest star beyond the Sun (Proxima Centauri) is 4.2 light-years away. One light-year is about 9.5 trillion km. If you built a spaceship that travels at 1% of the speed of light (about 3,000 km/s), the journey would take 420 years — far longer than a human lifetime. How would you design a spaceship for a multi-generational journey? What would the people on board eat, breathe, and do for 14 generations?

Questions to Ponder

Think Deeper

1. Different civilizations saw different patterns in the same stars. If you could create your own constellation, what shape would you connect and what story would you tell? Which stars would you include?
2. The nearest star beyond the Sun (Proxima Centauri) is 4.2 light-years away. If we built a spaceship travelling at 1% of the speed of light, the journey would take over 400 years. How would you design a spaceship for a journey that lasts many human generations? What would the people on board eat, breathe, and do?
3. We see the Milky Way as a hazy band across the sky because we are inside it. If you could snap your fingers and instantly view our galaxy from 10 million light-years away, what do you think it would look like? Draw what you imagine.
4. A red supergiant like Betelgeuse could explode as a supernova any day (or any thousand years from now). When it does, it will be bright enough to see during the daytime! Would you want to witness a supernova? What do you think ancient peoples thought when they saw a “new star” suddenly appear?
5. Light pollution hides most of the stars from city dwellers. If you were the mayor of a big city, what rules would you introduce to reduce light pollution while keeping the city safe at night?

What You Learned

Things this chapter taught you

- A **star** is a massive sphere of glowing plasma that produces light and heat through **nuclear fusion** in its core.
- A star's **colour** reveals its surface temperature: blue stars are hottest; red stars are coolest.

- **Constellations** are 88 officially recognized patterns used to map the celestial sphere.
- Our **Milky Way** is a barred spiral galaxy containing over 100 billion stars.
- **Light pollution** from cities obscures our view of the night sky.
- Tools like **planispheres** and pointer stars help us navigate the stars with the unaided eye.



WE live on a rocky planet that orbits an ordinary yellow star. That star is just one of hundreds of billions swirling within the Milky Way galaxy. And our Milky Way is merely a single galaxy among trillions scattered across the vast, expanding ocean of the universe. Let us venture beyond our solar system to explore the grandest scale of the cosmos and uncover how it all began.

What lies beyond our solar system, and how did the universe begin?

6.1

Our Galaxy: The Milky Way

The Milky Way is a barred spiral galaxy—a colossal, spinning disc composed of stars, gas, and interstellar dust bound together by gravity. It is so immense that a beam of light, traveling at the universe’s ultimate speed limit (300,000 km/s), would take 100,000 years just to cross from one edge of the galactic disc to the other!

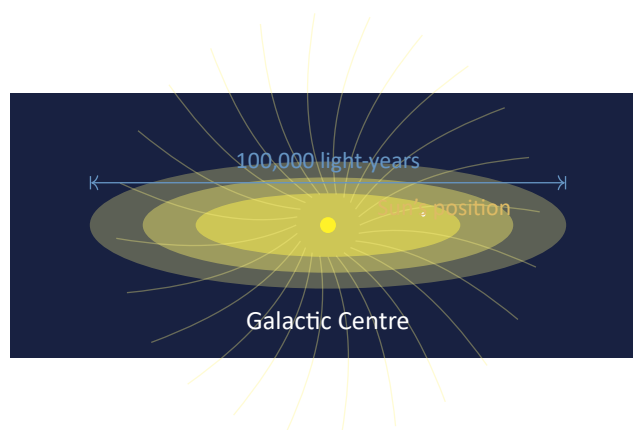


Figure 6.1: A profile view of the Milky Way. Our Sun resides safely in the galactic suburbs, roughly halfway out from the core.

Light-Year

A light-year is a unit of *distance*, not time! It is the distance light travels through a vacuum in one year—approximately 9.5 trillion kilometres. When astronomers observe a star that is 10 light-years away, they are seeing that star exactly as it looked 10 years ago.

The Galactic Year

Just as Earth orbits the Sun, our entire solar system orbits the centre of the Milky Way at roughly 828,000 km/h. Even at that incredible speed, it takes our Sun about 230 million years to complete one full circuit—a single “galactic year.” The last time our solar system was in its current position, the first dinosaurs were just beginning to walk the Earth!

What Is at the Centre?

At the gravitational heart of the Milky Way lies a region of intense energy and density. Hidden there is a **supermassive black hole** known as Sagittarius A* (pronounced “Sagittarius A-star”), packing a mass equivalent to 4 million Suns into a relatively tiny space.

Despite popular depictions in science fiction, black holes do not act like cosmic vacuum cleaners reaching out to indiscriminately suck in the galaxy. At a safe distance, a black hole’s gravity behaves just like that of any other object with the same mass—which is why millions of stars orbit harmlessly around Sagittarius A* like planets around a star.

6.2**Other Galaxies and the Cosmic Neighborhood**

The Milky Way is far from alone in the void. It is the second-largest member of a gravitational family known as the **Local Group**, a collection of more than 50 galaxies. Our largest galactic neighbor is the spiral Andromeda Galaxy, located about 2.5 million light-years away.

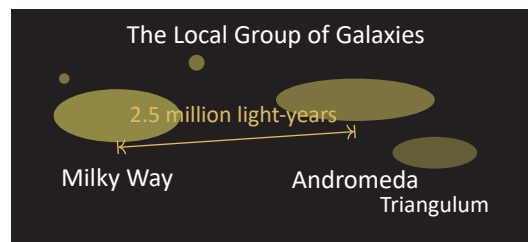


Figure 6.2: The Local Group is our cosmic neighborhood, dominated by the Milky Way and the Andromeda Galaxy.

Across the universe, galaxies come in three primary architectural forms:

- **Spiral galaxies** — Features majestic, flat, rotating discs with sweeping pinwheel arms rich in gas, dust, and newborn stars (like the Milky Way and Andromeda).
- **Elliptical galaxies** — Smooth, egg-shaped or spherical collections of older stars with very little interstellar dust or gas to form new stars.

- **Irregular galaxies** — Systems lacking a distinct symmetrical shape, often distorted or torn apart by the gravitational pull of nearby larger galaxies or past cosmic collisions.

When Galaxies Collide!

While the universe at large is expanding, gravity dominates on a local scale. The Milky Way and Andromeda are currently racing toward one another at about 110 km/s. In roughly 4.5 billion years, they will collide and merge to form a single giant elliptical galaxy! However, because the distances between individual stars are so immense, stellar collisions will be virtually non-existent—our solar system will simply be redirected into a new galactic orbit.

6.3

The Expanding Universe

In the 1920s, American astronomer Edwin Hubble made one of the most profound discoveries in human history: distant galaxies are moving away from us! By analyzing the light emitted by these galaxies, he realized a pattern: the farther away a galaxy is located, the faster it is receding into the distance. This observation led to an astonishing conclusion: the fabric of the universe itself is **expanding**.

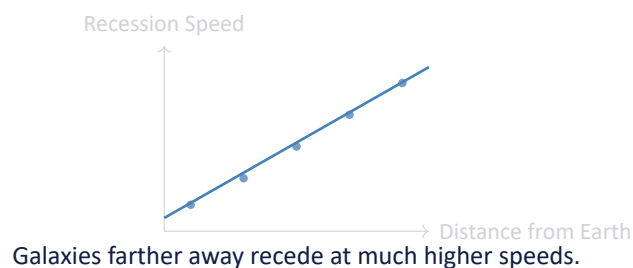


Figure 6.3: Hubble's Law: The direct relationship between a galaxy's distance and the speed at which it moves away from us.

To picture this, imagine a loaf of raisin bread dough rising in an oven. As the dough expands, every raisin moves away from every other raisin. If you were sitting on any single raisin, it would look like all the other raisins were rushing away from you! The universe behaves in much the same way—galaxies are not flying *through* empty, static space; rather, the physical space *between* the galaxies is continuously stretching and expanding.

The Hubble Space Telescope

Named in honor of Edwin Hubble, the Hubble Space Telescope was deployed into low Earth orbit in 1990. Operating above the blurring effects of Earth's atmosphere, it has captured the most iconic images of deep space ever recorded. Its famous "Hubble Ultra Deep Field" image revealed thousands of distant, colorful galaxies teeming within a patch of sky no larger than a grain of sand held at arm's length!

6.4

The Beginning: The Big Bang

If the universe is expanding today, then logical deduction tells us that in the past, everything must have been closer together. If you rewind the cosmic clock far enough—about 13.8 billion years—all the matter, energy, and space in the entire observable universe was compressed into an unimaginably dense, hot point. The rapid expansion from this initial state is known as the **Big Bang**.

It Was Not a Bomb!

A common misconception is that the Big Bang was an explosion of matter erupting *into* pre-existing empty space. In reality, the Big Bang was the sudden, rapid expansion of space and time themselves! There was no “outside” to explode into, and asking what happened “before” the Big Bang is like asking what lies north of the North Pole—time itself began with this cosmic event.

Three Pillars of Evidence for the Big Bang

Why do scientists accept the Big Bang theory? It is supported by three major lines of observational evidence:

1. **The Expansion of Space:** Edwin Hubble’s observation that galaxies are universally receding from one another confirms that the cosmos is actively growing.
2. **The Cosmic Microwave Background (CMB):** If the early universe was intensely hot and dense, it should have left behind a cooled-down glow of radiation. Astronomers detect this faint, ambient microwave energy coming uniformly from every direction in the sky—the literal cooling afterglow of creation!
3. **The Abundance of Light Elements:** The physics of the Big Bang predicts that the universe should be composed of roughly 75% hydrogen and 25% helium (by mass), with tiny traces of lithium. This matches exactly what we observe when we analyze the oldest stars and gas clouds in the universe.

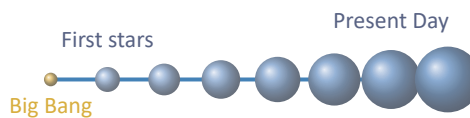


Figure 6.4: A timeline of cosmic evolution: from the initial singularity of the Big Bang to the sprawling, modern universe.

CMB: The Oldest Baby Picture

Discovered by accident in 1965 by radio astronomers Arno Penzias and Robert Wilson, the Cosmic Microwave Background (CMB) is the oldest light we will ever be able to see. It originated just 380,000 years after the Big Bang, at the exact moment the universe cooled

enough for electrons and protons to form neutral atoms, allowing light to finally travel freely through space!

6.5

The Mystery Ingredients: Dark Matter and Dark Energy

Perhaps the most humbling discovery of modern astronomy is that everything we have ever observed with our eyes and telescopes—every star, planet, comet, moon, gas cloud, and living creature—accounts for just $\sim 5\%$ of the total mass and energy content of the universe. The remaining 95% consists of two invisible, mysterious components.

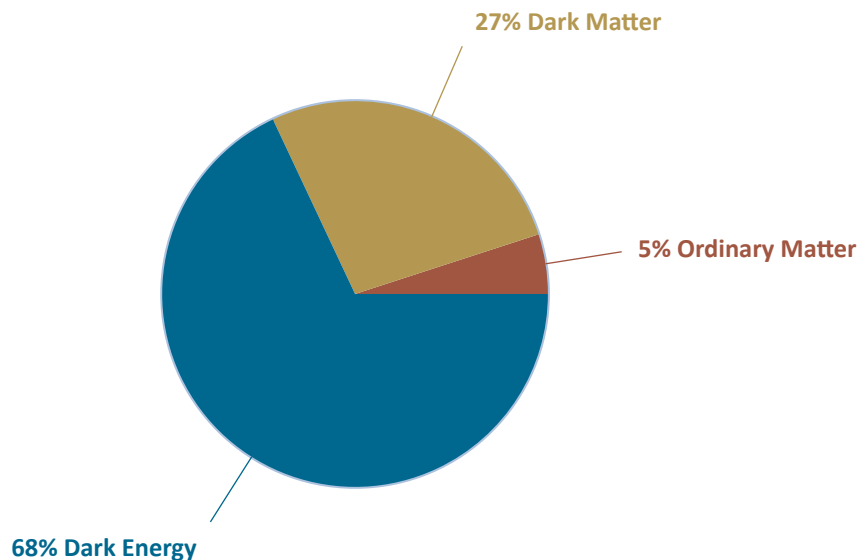


Figure 6.5: The cosmic recipe: ordinary, visible matter represents only a tiny slice of what makes up our universe.

Dark Matter (27%) An invisible form of matter that does not emit, absorb, or reflect any electromagnetic radiation (light), making it completely transparent. However, we know it exists because of its gravitational pull. Without the extra gravitational “glue” provided by dark matter, spinning galaxies like the Milky Way would fly apart into space!

Dark Energy (68%) A mysterious, repulsive force permeating all of empty space that is causing the expansion of the universe to accelerate over time. While dark matter acts as a cosmic brake pulling galaxies together, dark energy acts as a cosmic accelerator driving them apart faster and faster.

A Universe of Mysteries

Scientists estimate that 95% of our universe is made of dark matter and dark energy—substances we cannot see or detect with traditional light-based telescopes. How do you study something that is invisible? If you were an astrophysicist, what kind of indirect

experiment or creative instrument would you design to prove the existence of an invisible particle or force?

Find the Andromeda Galaxy with Your Eyes

On a clear, moonless autumn or winter night, find the constellation Cassiopeia (which looks like a W or M in the sky). Follow the point of the W toward the next bright region — that is the Andromeda Galaxy (M31)!

Observe: To the naked eye, Andromeda looks like a faint, fuzzy patch of light. But that faint smudge is actually a trillion stars — an entire galaxy 2.5 million light-years away! You are seeing light that began its journey before humans existed. Try with binoculars for a much better view.

Sort Galaxies Like Hubble

Visit the [Galaxy Zoo](#) website (with a parent's permission) and try classifying real galaxies from telescope images. Scientists need help identifying spiral, elliptical, and irregular galaxies from millions of images!

Offline version: Search online for “galaxy classification images” and print a few. Sort them into spiral, barred spiral, elliptical, and irregular categories using what you learned in this chapter. Check your answers against the official classifications.

Make Your Own Expanding Universe

You can model the expansion of space using a simple rubber balloon! Take a deflated balloon and use a permanent marker to draw several dots across its surface (each dot represents a galaxy). Slowly blow up the balloon and observe what happens to the dots. Notice that no matter which dot you pick as your “home galaxy,” all the other dots move away from it. Notice also that dots located farther apart separate much faster than dots located close together—just like Hubble's Law!

6.6

Chapter Review

Key Concepts

- The **Milky Way** is a barred spiral galaxy containing hundreds of billions of stars, with a supermassive black hole at its centre.
- A **light-year** is a unit of distance representing how far light travels through a vacuum in one year (~ 9.5 trillion km).
- Edwin Hubble discovered that distant galaxies are moving away from us, proving that the **universe is expanding**.

- The **Big Bang theory** explains that space, time, matter, and energy began expanding from a hot, dense state 13.8 billion years ago.
- The observable universe is primarily composed of invisible **dark energy** (68%) and **dark matter** (27%), with ordinary atomic matter making up only about 5%.

Review Questions

1. What type of galaxy is the Milky Way, and roughly how long would it take a beam of light to cross its diameter?
2. Why is the term “light-year” slightly confusing to people who are new to astronomy? What does it actually measure?
3. What did Edwin Hubble notice about the light coming from distant galaxies, and what did this tell us about the fabric of space?
4. Why is it incorrect to describe the Big Bang as an explosion of matter erupting into empty space?
5. What is the Cosmic Microwave Background (CMB), and why is it considered a pillar of evidence for the Big Bang?
6. What observation forces astronomers to conclude that invisible “dark matter” must exist inside spinning galaxies?

Words to Know

Big Bang The primordial event 13.8 billion years ago that marks the beginning of the expansion of space, time, matter, and energy.

Black Hole A region of space where gravity is so intense that nothing—not even light—can escape its boundary (the event horizon).

Cosmic Microwave Background (CMB) The faint, cooled-down remnant radiation left over from the early, hot stages of the Big Bang that fills the entire universe.

Dark Energy A mysterious, repulsive form of energy inherent to empty space that is driving the accelerated expansion of the universe.

Dark Matter An invisible, non-luminous form of matter that interacts with ordinary matter primarily through its gravitational pull.

Light-Year The astronomical distance that a photon of light travels in a vacuum over the course of one Earth year (~ 9.5 trillion km).

Local Group The gravitationally bound cluster of more than 50 galaxies that includes the Milky Way, Andromeda, and the Triangulum Galaxy.

Supernova The colossal, cataclysmic explosion of a massive star at the end of its stellar lifecycle, forging and scattering heavy elements across a galaxy.

Things to do!

Observe

Find the Andromeda Galaxy with your naked eye! On a clear, moonless autumn or winter night, find Cassiopeia (a W shape in the sky). Follow the point of the W toward the next fuzzy patch. That faint smudge is the Andromeda Galaxy — a trillion stars, 2.5 million light-years away. The light entering your eyes left Andromeda before humans existed!

Think

When you look at the Andromeda Galaxy, you see it as it was 2.5 million years ago. If an alien in Andromeda pointed a super-telescope at Earth right now, they would see a planet with no humans — they would see early Ice Age Earth. Where in the universe would you need to be to see the dinosaurs? Where would you need to be to see the Big Bang?

Try It

Model the expanding universe with a balloon! Draw dots on a deflated balloon (each dot = a galaxy). Inflate it slowly. Watch every dot move away from every other dot. Notice that dots farther apart separate faster — this is exactly Hubble's Law! No matter which dot you call home, all other dots recede. You have just modelled the expanding universe with a party balloon.

Explore Further

- **James Webb Space Telescope Images:** <https://webbtelescope.org>
- **Hubble Space Telescope Deep Field:** <https://hubblesite.org>
- **Galaxy Zoo — Classify Real Galaxies:** <https://galaxyzoo.org>
- **ESA Kids — Big Bang Timeline:** <https://www.esa.int/kids>
- **NASA Space Place — The Universe:** <https://spaceplace.nasa.gov/universe/>
- **CERN — The Mystery of Dark Matter:** <https://home.cern/science/physics/dark-matter>

★ Did You Know?

The Cosmic Microwave Background (CMB) is the oldest light in the universe. It is the faint afterglow of the Big Bang, emitted when the universe was only 380,000 years old. This radiation fills all of space and is only 2.7 degrees above absolute zero. If your TV is tuned to a channel with no signal, about 1% of the static you see is light from the Big Bang!

 Challenge

Dark matter makes up 27% of the universe, but we have never directly seen it. We know it exists because galaxies spin too fast to hold together without extra gravity. If you were an astrophysicist, how would you design an experiment to detect invisible particles? Where would you build your detector, and what would you look for?

Questions to Ponder

Think Deeper

1. When you look at the Andromeda Galaxy, you see it as it was 2.5 million years ago. If an alien in Andromeda looked at Earth right now, what would they see? (Hint: humans did not exist 2.5 million years ago.) Does this change how you think about “now”?
2. The Milky Way and Andromeda will collide in about 4.5 billion years. By then, the Sun will be running out of fuel and Earth may be uninhabitable. If humanity survives that long, where should we go? What would you want to save from Earth?
3. Dark matter makes up 27% of the universe, but we have never directly detected it. How confident are you that it exists? What would it take to convince you that dark matter is real, or that we might be wrong?
4. The universe is expanding and the expansion is accelerating. If you could fast-forward trillions of years, all galaxies beyond our Local Group will have moved so far away that they become invisible. Future astronomers might look up at an empty, black sky. What would they conclude about the universe?
5. The Big Bang created time and space itself. What do you think “before” the Big Bang means? Is it possible to imagine a state without time? What questions would you ask a theoretical physicist about this?

What You Learned

Things this chapter taught you

- The **Milky Way** is a barred spiral galaxy with a supermassive black hole at its centre.
- A **light-year** is the distance light travels in one year — about 9.5 trillion kilometres.
- Edwin Hubble discovered that the **universe is expanding**: distant galaxies move away faster (Hubble’s Law).
- The **Big Bang** started the expansion of space, time, and matter 13.8 billion years ago.

- The **CMB** is the faint afterglow of the Big Bang, filling all of space.
- **Dark matter** (27%) and **dark energy** (68%) dominate the universe; ordinary matter is only 5%.
- A **black hole** is a region where gravity is so strong that nothing, not even light, can escape.
- A **supernova** is the explosive death of a massive star, scattering heavy elements across space.

★ ★ ★

1. Planet Quick Reference

Planet	Distance	Diameter	Day Length	Year Length	Moons
	(million km)	(km)			
Mercury	57.9	4,879	59 Earth days	88 days	0
Venus	108.2	12,104	243 Earth days	225 days	0
Earth	149.6	12,742	24 hours	365.25 days	1
Mars	227.9	6,779	24.6 hours	687 days	0
Jupiter	778.5	139,820	9.9 hours	11.9 years	95
Saturn	1,434	116,460	10.7 hours	29.5 years	146
Uranus	2,871	50,724	17.2 hours	84 years	28
Neptune	4,495	49,244	16.1 hours	165 years	16

title=Quick Classifications

- **Inner Planets (Terrestrial):** Mercury, Venus, Earth, Mars — Small, dense, and rocky.
- **Outer Planets (Gas & Ice Giants):** Jupiter, Saturn, Uranus, Neptune — Large, gaseous/icy, with extensive moon systems.

2. The Greek Alphabet

Astronomers use Greek letters to designate star brightness rankings within a constellation (e.g., α Centauri).

Name	Upper	Lower	Name	Upper	Lower
Alpha	A	α	Nu	N	ν
Beta	B	β	Xi	Ξ	ξ
Gamma	Γ	γ	Omicron	O	o
Delta	Δ	δ	Pi	Π	π
Epsilon	E	ϵ	Rho	P	ρ
Zeta	Z	ζ	Sigma	Σ	σ
Eta	H	η	Tau	T	τ
Theta	Θ	θ	Upsilon	Υ	υ
Iota	I	ι	Phi	Φ	ϕ
Kappa	K	κ	Chi	X	χ
Lambda	Λ	λ	Psi	Ψ	ψ
Mu	M	μ	Omega	Ω	ω

3. Astronomical Constants & Metric Prefixes

Quantity	Symbol	Standard Value
Speed of Light	c	3.00×10^8 m/s (300,000 km/s)
Gravitational Constant	G	6.67×10^{-11} N · m ² /kg ²
Astronomical Unit	AU	1.50×10^{11} m (Average Earth–Sun distance)
Light-Year	ly	9.46×10^{15} m (Distance light travels in 1 year)
Parsec	pc	3.09×10^{16} m $\approx$ 3.26 ly
Solar Mass	$M_{\odot}$	1.99×10^{30} kg
Solar Radius	$R_{\odot}$	6.96×10^8 m
Earth Mass	$M_{\oplus}$	5.97×10^{24} kg
Earth Radius	$R_{\oplus}$	6.37×10^6 m
Hubble Constant	H_0	$\approx$ 70 km/s/Mpc

Prefix	Symbol	Notation	Factor
Giga	G	10^9	1 Billion
Mega	M	10^6	1 Million
Kilo	k	10^3	1 Thousand
Centi	c	10^{-2}	1 Hundredth
Milli	m	10^{-3}	1 Thousandth
Micro	μ	10^{-6}	1 Millionth
Nano	n	10^{-9}	1 Billionth

4. Key Equations for Stargazers

title=Fundamental Observational & Physical Formulas

Stellar Parallax (Distance)

$$d = \frac{1}{p}$$

Symbols: d = distance in parsecs (pc), p = parallax angle in arcseconds (").

Pogson's Formula (Brightness Ratio)

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right)$$

Symbols: m_1, m_2 = apparent magnitudes, F_1, F_2 = radiant flux (brightness in W/m²).

Distance Modulus

$$m - M = 5 \log_{10}(d) - 5$$

Symbols: m = apparent magnitude, M = absolute magnitude, d = distance in pc.

Kepler's Third Law

$$P^2 = a^3$$

Symbols: P = orbital period (Earth years), a = semi-major axis (AU).

Wien's Displacement Law

$$\lambda_{\max} = \frac{b}{T}$$

Symbols: $\lambda_{\max}$ = peak wavelength (m), T = surface temperature (K), $b \approx 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$.

Stefan-Boltzmann Law (Luminosity)

$$L = 4\pi R^2 \sigma T^4$$

Symbols: L = luminosity (W), R = radius (m), T = temperature (K), $\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$.

5. Example & Practice Problems

Level 1

title=Example Problem: How Long Does Sunlight Take to Reach Earth?

Question: Light travels at 300,000 km/s, and the Sun is about 150,000,000 km away from Earth. How many minutes does it take for sunlight to reach us?

Solution:

1. Time in seconds = $\frac{\text{Distance}}{\text{Speed}} = \frac{150,000,000 \text{ km}}{300,000 \text{ km/s}} = 500 \text{ seconds.}$
2. Convert to minutes: $\frac{500}{60} \approx 8.33 \text{ minutes (or 8 minutes and 20 seconds).}$

title=Practice Problem 1.1 (Elementary)

Problem: Jupiter's diameter is roughly 11 times larger than Earth's diameter (12,742 km). Estimate Jupiter's diameter and compare it to the actual value in Section 1.

Level 2

title=Example Problem: Would Saturn Float in Water?

Question: Saturn has a mass of $M = 5.68 \times 10^{26} \text{ kg}$ and a volume of $V = 8.27 \times 10^{23} \text{ m}^3$. Calculate its average density. Would it float in water ($\rho_{\text{water}} = 1,000 \text{ kg/m}^3$)?

Solution:

1. Density formula: $\rho = \frac{M}{V} = \frac{5.68 \times 10^{26} \text{ kg}}{8.27 \times 10^{23} \text{ m}^3} \approx 687 \text{ kg/m}^3.$
2. Since $687 \text{ kg/m}^3 < 1,000 \text{ kg/m}^3$, Saturn's average density is less than water, so it would float!

title=Practice Problem 2.1 (Middle School)

Problem: An asteroid orbits the Sun at an average distance of $a = 4 \text{ AU}$. Using Kepler's Third Law ($P^2 = a^3$), calculate how many Earth years it takes to complete one orbit around the Sun.

Level 3

Note for younger readers: Level 3 introduces algebra with logarithms — a topic usually taught in later high school. Do not worry if these problems feel challenging! Simply read through the worked example to build intuition, and come back to the practice problem when you are ready.

title=Example Problem: Stellar Distance via Distance Modulus

Question: A star has an apparent magnitude of $m = 8.5$ and an absolute magnitude of $M = 3.5$. Calculate its distance in parsecs.

Solution:

1. Distance modulus equation: $m - M = 5 \log_{10}(d) - 5$
2. Substitute values: $8.5 - 3.5 = 5 = 5 \log_{10}(d) - 5 \implies 10 = 5 \log_{10}(d)$
3. Divide by 5: $\log_{10}(d) = 2 \implies d = 10^2 = 100 \text{ parsecs.}$

title=Practice Problem 3.1 (High School / Competition)

Problem: Star Betelgeuse has a surface temperature of approximately $T = 3,500$ K. Use Wien's Displacement Law ($\lambda_{\text{max}} = b/T$) to calculate its peak emission wavelength and determine which region of the spectrum it falls in.

6. Fundamental Open-Ended Questions for Young Stargazers

title=Questions to Explore & Discuss

1. **Olbers' Paradox:** If the universe has billions of glowing stars, why is the night sky dark instead of blazing bright white?
2. **The Mystery of Orbiting:** If gravity constantly pulls the Moon toward Earth, why doesn't the Moon crash directly into us?
3. **Stellar Fingerprints:** We cannot travel to distant stars, so how do astronomers know exactly what chemical elements make up a star billions of miles away?
4. **Flat Solar System:** Why do almost all planets in our Solar System orbit the Sun in a flat plane, like marbles rolling on a dinner plate?
5. **Two Moons on Earth:** What would happen to Earth's ocean tides, climate, and night sky if Earth had two moons instead of one?
6. **The Speed of Gravity:** If the Sun suddenly vanished, would Earth fly off into space immediately, or would it take a few minutes? How would we test this?
7. **Black Hole Horizon:** If you could watch someone fall into a black hole, they would appear to slow down and freeze at the event horizon. But from their perspective, they cross it normally. Which viewpoint is correct?
8. **Alien Astronomers:** If an alien civilization 65 million light-years away pointed a powerful telescope at Earth right now, what would they see? Does this mean the past is still visible somewhere out there?
9. **Faster Than Light?** Einstein says nothing can travel faster than light. But the universe itself expanded faster than light during inflation. Is this a loophole? What counts as "travel" through space?
10. **The Fermi Paradox:** The universe is vast and old. Statistically, alien civilizations should exist. So where is everybody? What are some possible explanations for why we have not heard from anyone?

6a. Hands-On Activities

Stargazing Journal

Start a nightly astronomy journal. Each clear night, go outside and write down:

- Date, time, and weather conditions
- Moon phase (draw it!)
- Brightest stars or planets visible
- Any unusual observations (shooting stars, satellites, etc.)

Challenge: After one month, look back at your entries. Can you predict the Moon's phase for tomorrow? Can you identify which bright dots are planets (they don't twinkle!) versus stars?

Scale Model of the Solar System

Find a long stretch of space (a football field or long hallway). Use a scale where the Sun is a grapefruit (15 cm diameter). Using this scale:

- Earth would be a tiny peppercorn about 16 metres away
- Jupiter would be a marble about 82 metres away
- Pluto (dwarf planet) would be a grain of sand about 630 metres away!

Observe: Walk the distances yourself. The vast empty spaces between planets are hard to imagine from diagrams in books. Space is truly mostly empty!

Pinhole Camera to View the Sun

NEVER look at the Sun directly. Instead, make a pinhole projector:

1. Take two pieces of cardboard.
2. Poke a small, clean hole in the centre of one piece.
3. Stand with your back to the Sun.
4. Hold the pierced cardboard above the other piece.
5. Adjust the distance until you see a clear image of the Sun on the lower card.

Observe: During a solar eclipse, you can see the Moon's silhouette crossing the Sun's face using this safe method! The image will be upside down because light travels in straight lines through the hole.

Crater Creator Simulation

Explore how meteorites shape planetary surfaces like the Moon and Mercury:

1. Fill a shallow baking dish with 2–3 cm of flour.
2. Dust a thin layer of cocoa powder evenly over the surface.
3. Drop small objects (marbles, small rocks, steel bearings) into the dish from different heights.

Observe: Notice the "ejecta rays"—the lighter flour thrown outward from the impact center. How does changing the drop height or object size affect the crater's diameter and depth?

Balloon Universe Model

Demonstrate how space itself expands, pushing distant galaxies farther apart:

1. Partially inflate a round balloon (do not tie it yet).
2. Use a permanent marker to draw 5–6 dots across the balloon to represent galaxies.
3. Choose one "home" galaxy and measure the distance to the others with a flexible ruler or string.
4. Blow up the balloon fully and re-measure the distances.

Observe: Notice that galaxies aren't moving *through* space; rather, space itself is expanding between them. The farther away a galaxy is from your home galaxy, the faster it appears to recede!

Handheld Stellar Parallax

Understand how astronomers measure distances to nearby stars using apparent position shifts:

1. Hold your thumb upright at arm's length against a distant background (a wall or tree line).
2. Close your left eye and look at your thumb with your right eye.
3. Switch eyes rapidly (close right, open left) without moving your hand.

4. Bring your thumb closer to your face (about 15 cm) and repeat the eye-switching.

Observe: Your thumb appears to jump back and forth against the background. When your thumb is closer to your eyes, the shift is dramatically larger. Astronomers use this exact line-of-sight shift to measure distances to nearby stars as Earth orbits the Sun.

Constellation Star Count

Assess the impact of light pollution on your local viewing conditions:

- Locate a well-known constellation (such as Orion or Cygnus).
- Step outside on a clear night and allow your eyes 15 minutes to adjust to the dark.
- Count every individual star you can spot *inside* the main pattern or outline.
- Compare your counts from different locations (e.g., your backyard, a city park, or a rural dark site).

Challenge: Under a pristine dark sky, you can spot dozens of faint stars within Orion's box. How many can you count from your neighbourhood? Use your count to estimate your local sky quality on the Bortle Scale.

Modeling Moon Phases

Understand why the Moon's shape appears to change throughout the month:

1. Place a single bright lamp in the centre of a dark room (representing the Sun).
2. Hold a small white foam ball at arm's length (representing the Moon).
3. Your head represents Earth. Slowly rotate in place while holding the ball at arm's length.

Observe: Watch how the lit half of the ball changes from a thin crescent to a full circle as you spin. Notice that the Moon is always half-lit by the Sun; its apparent shape changes entirely based on your viewing angle!

DIY Cereal Box Spectroscope

Astronomers analyse starlight by splitting light into a rainbow. Make your own spectroscope:

1. Cut a thin vertical slit in one side of an empty cereal box.
2. Cut a small viewing hole on the top edge at a 45° angle.
3. Insert an old CD or DVD inside the box opposite the slit so light bounces off it.
4. Point the slit at safe light sources (household lamps, LEDs, fluorescent lights—never the Sun!).

Observe: Look through the viewing hole to see rainbow emission spectra! Fluorescent lights produce sharp individual lines, while incandescent bulbs produce a smooth, continuous spectrum.

Apparent Retrograde Motion

Demonstrate why planets like Mars occasionally appear to move backward across the night sky:

1. Draw two concentric circles on the ground: an inner track (Earth) and an outer track (Mars).
2. Have two people walk counterclockwise around the tracks, with the inner walker moving faster.
3. Have the inner walker track the outer walker's position against a distant fence or wall.

Observe: As the inner walker laps the outer walker, the outer person temporarily appears to drift backward against the background. This perspective effect explains planetary retrograde motion!

Flashlight Constellation Projector

Bring the night sky indoors by building a simple star projector:

1. Cut out circles of aluminium foil or heavy black paper to fit over the end of a flashlight.
2. Use a pushpin to punch hole patterns matching real constellations (e.g., Ursa Major, Cassiopeia).
3. Secure the foil disk over the front of the flashlight using a rubber band.
4. Turn off the lights and project the stars onto a dark ceiling or wall.

Challenge: Try creating several constellation disks and see if family members can identify the star patterns on the ceiling!

Greenhouse Effect in a Bottle

Explore why Venus is hotter than Mercury, despite being farther from the Sun:

1. Place a thermometer inside two identical clear plastic bottles.
2. Leave one bottle unsealed, and seal the cap tightly on the second bottle.
3. Place both bottles in direct sunlight or under a bright heat lamp for 20 minutes.
4. Record the temperature in both bottles every 5 minutes.

Observe: The sealed bottle traps heat inside, driving the temperature significantly higher than the open bottle. Venus's dense atmosphere traps solar heat in the exact same way, causing an extreme runaway greenhouse effect.

7. Problem Archives & Practice Resources

Resource / Organization	Description & Web Link
Bangladesh Astronomy Olympiad (BDOAA)	National olympiad portal featuring past papers, regional question banks, and competition prep material. Website: https://bdoaa.org/
International Olympiad (IOAA)	Complete collection of theory, observation, and data analysis exam problems from global competitions. Website: https://www.ioaastrophysics.org/
USA Astronomy Olympiad (USAAAO)	Free archive of preliminary and national exam papers complete with step-by-step solution keys. Website: https://usaaao.org/
International Astronomy Olympiad (IAO)	Official problems and solution sets from long-running international astronomy competitions. Website: http://www.issp.ac.ru/iao/
NASA JPL Education	Hands-on math and space problem sets ("Pi in the Sky" and "Teachable Moments") for students and educators. Website: https://www.jpl.nasa.gov/edu/
American Association of Variable Star Observers (AAVSO)	Real light curve data sets and magnitude calculation activities for amateur astronomers. Website: https://www.aavso.org/
MIT OpenCourseWare	Open university problem sets and lecture notes from introductory astrophysics courses (Course 8.282J). Website: https://ocw.mit.edu/

8. Sun & Moon Comparison

Property	The Sun	The Moon
Mass	1.99×10^{30} kg	7.34×10^{22} kg
Radius	696,000 km	1,740 km
Surface Temperature	5,500°C	−180°C to 120°C
Distance from Earth	~ 150 million km	384,400 km
Age	~ 4.6 billion years	~ 4.5 billion years

Data compiled from NASA/JPL Data Archives.



A JOURNEY THROUGH THE UNIVERSE



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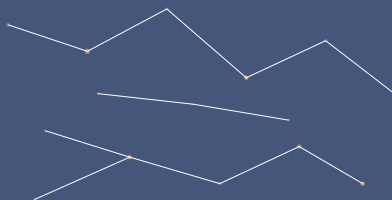
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